School of Computing and Artificial Intelligence
Faculty of Engineering and Technology
Sunway University

SYSTEM TEST DOCUMENT

SEMESTER	: JULY 2025
COURSE NAME	: BIS2102 INFORMATION SYSTEM ANALYSIS AND DESIGN
INSTRUCTOR	: ASSOC. PROF. TS. DR. ASLINA BAHARUM
SYSTEM NAME	: Booking.com System
PROGRAM NAME	: Bachelor of Information Systems (Data Analytics)
CONTACT PERSON	: LIANG YRENN DA 22067128@imail.sunway.edu.my

STUDENT ID	STUDENT NAME	GROUP	ROLE
22067128	**LIANG YRENN DA**	BIS 1-4	Project Leader
22071112	**YEOH JIE LONG**	BIS 1-4	System Analyst
22039069	**OOI JIN YON**	BIS 1-4	UI Coordinator
25018565	**WONG XU LIANG**	BIS 1-4	Tester / QA Lead
24082968	**MUHAMMAD HARIZ HAKIMI BIN MOHD HAFIZ**	BIS 1-4	UX Coordinator

DATE OF SUBMISSION	19/7/2025
ACTUAL DATE OF SUBMISSION	19/7/2025
NO OF DAYS LATE	0

Table of Contents

1.0 Use Case Testing 3
1.1 Tour Guide Booking 3
1.1.1 Test Case 1 3
1.1.2 Test Case 2 9
1.1.3 Test Case 3 15
1.1.4 Test Case 4 22
1.2 Manual Booking 29
1.2.1 Test Case 1 29
1.2.2 Test Case 2 31
1.2.3 Test Case 3 35
1.3 Transportation Booking 38
1.3.1 Test Case 1 38
1.3.2 Test Case 2 39
1.3.3 Test Case 3 42
1.3.4 Test Case 4 44
1.4 AI Trip Planner 47
1.4.1 Test Case 1 47
1.4.2 Test Case 2 49
1.4.3 Test Case 3 52
2.0 Initial Prototype 55
3.0 Testing Process 66
3.1 User Interface Design 66
3.1.1 Design a transparent interface 66
3.1.2 Create easy to learn and use interface 66
3.2 Pilot Test 70
3.3 Usability Testing 70
4.0 Final Prototype 97
5.0 Digital Mockup 101
6.0 Appendix 109

1.0 Use Case Testing

1.1 Tour Guide Booking

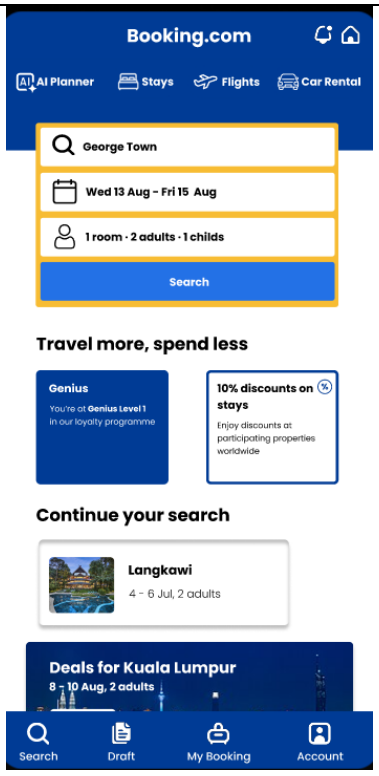
Tester	Wong Xu Liang
Test date	14 July 2025
Prototype developer	Liang Yrenn Da

1.1.1 Test Case 1

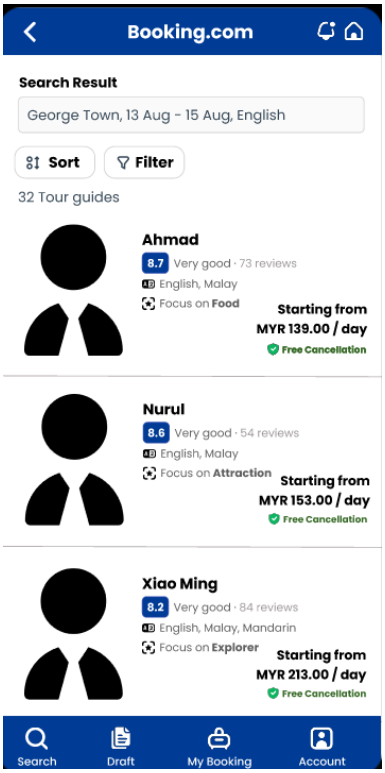
Test Objective	<u>Normal flow</u> To test the tour guide booking process
Potential Test Inputs (with example)	1. The user selects “Continue with email” as login method. 2. The user enters the email address. 2.1. Example: admin@gmail.com 3. The user enters the verification code 3.1. Example: 123456 4. The user navigates and clicks on the tour guide section 5. The user enters search details: example; 5.1. Destination: George Town 5.2. Duration: 13 August – 15 August 5.3. Attendees: 1 room for 2 adults and 1 child 6. The user clicks the search button 7. The user clicks on Ahmad’s profile 8. The user clicks Book Now 9. The user clicks Proceed to Payment 10. The user enters their card details for payment 11. The user clicks the pay button 12. The user clicks the “Get booking confirmation”
Expected Test Outputs (with example)	1. A prompt appears requesting the user enter their email address

	2. A prompt appears to enter the verification code sent to the provided email 3. The user is successfully logged in and redirected to the main dashboard/homepage 4. A list or interface for tour guide services is displayed 5. Search form accepts input for destination, duration, and attendees 6. Search results show available tour guides based on search details 7. Ahmad's tour guide profile opens with details, pricing, availability, and reviews 8. Booking summary page appears with trip details, price, and a "Proceed to Payment" option 9. Payment page opens with fields for card/payment details 10. Card fields are populated correctly; system validates card format 11. Payment is processed. A success message 12. A confirmation page is displayed with booking reference number, details of the tour, and possibly an option to download or email the confirmation
Test Procedures	1. Enter "Continue with email" button 2. Enter your email address and click continue 3. Refer to the verification code sent by the system, enter the code and click verify email 4. In the home page, navigate to the Tour Guide section using the header 5. Enter "George Town" for destination in the search tab

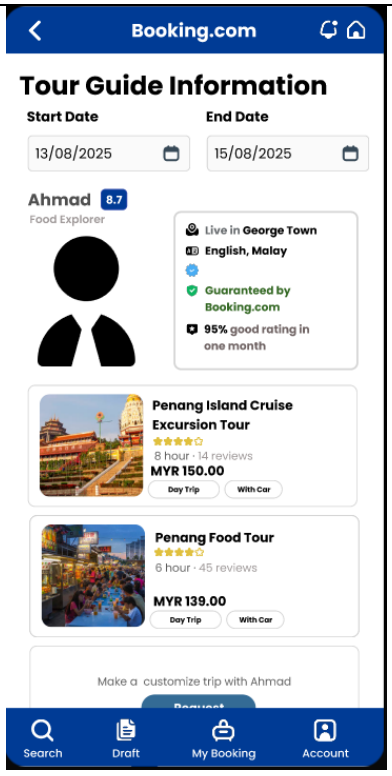
	6. Enter 13 August to 15 August for duration in the search tab 7. Enter 2 adults 1 child for number of attendees in the search tab 8. Enter English for language preference on tour guide in your search tab 9. Click search 10. Browse through the list of tour guide 11. Click on Ahmad tour guide 12. Browse through Ahmad's profile 13. Confirm itinerary by clicking "Book Now" button 14. Confirm total pricing and click "Proceed to Payment" 15. Select payment method "New card" 16. Enter your card details 17. Click "Pay MYR 289.00" for payment 18. Payment shows successfully booked the tour guide with payment details 19. Click on "Get boking confirmation" to proceed to confirmation page 20. Confirm that the tour guide booking displays on your booking page
Actual Test Results	<specify the actual outcome of the prototype> 1. Users are able to successfully login into their account and access homepage: **successful**



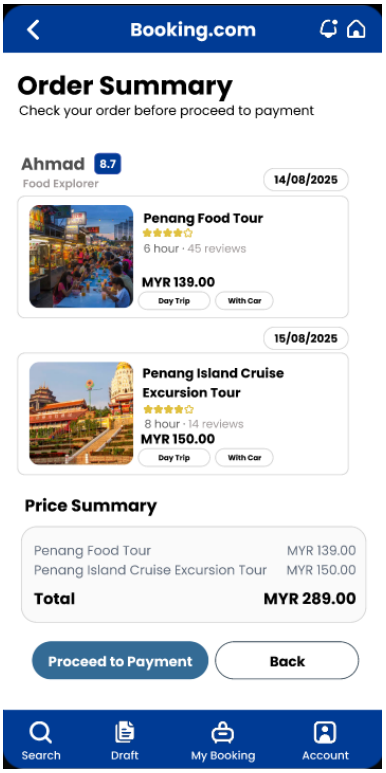
2. User is able to search up list of tour guides: **successful**



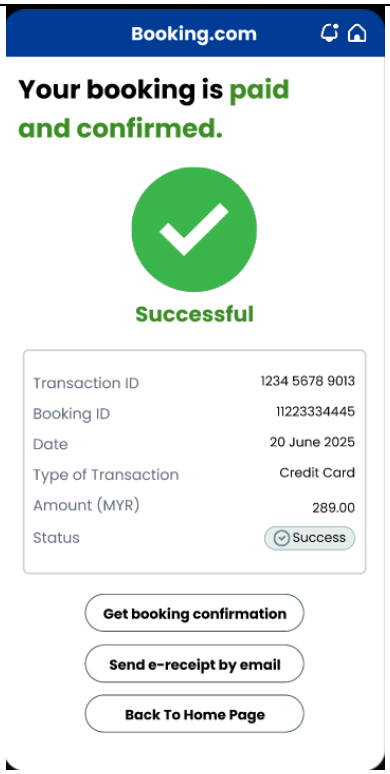
3. User is able to view tour guides infomation: **successful**



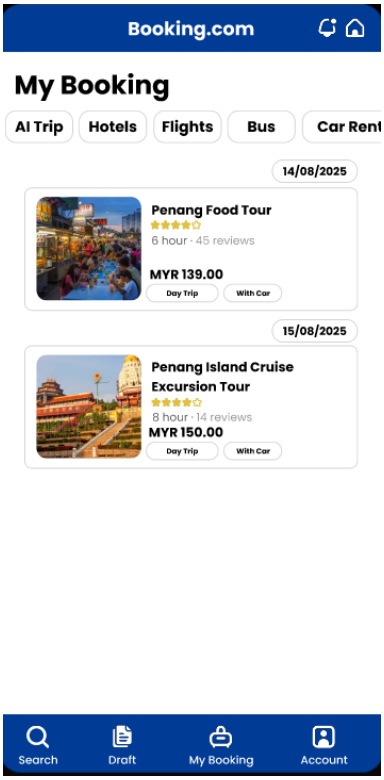
4. User is able to confirm booking with tour guide: **successful**



5. User is able to make payment: **successful**



6. User is able to view their booking confirmation: **successful**

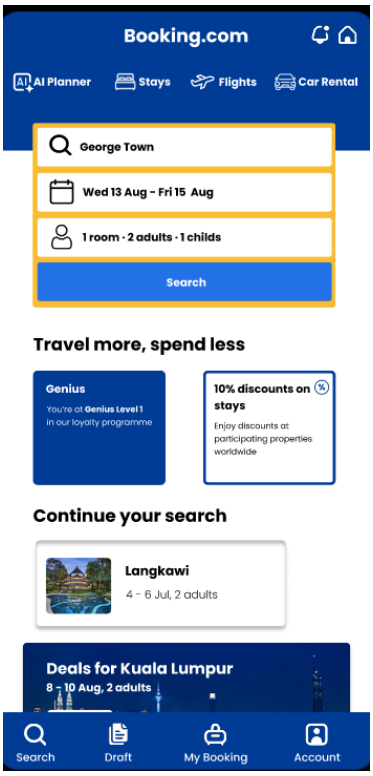
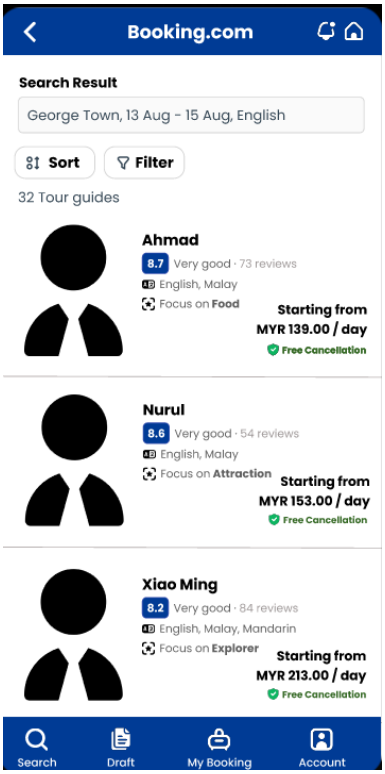


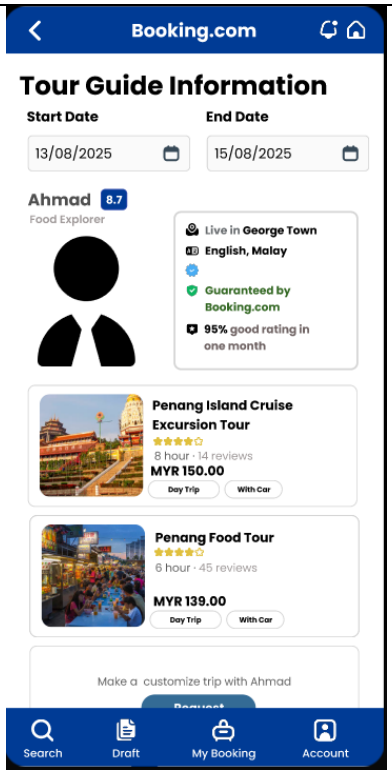
1.1.2 Test Case 2

Test Objective	Alternate flow Login unsuccessful when testing the tour guide booking process
Potential Test Inputs (with example)	1. The user selects “Continue with email” as login method. 2. The user enters the email address. 2.1. Example: admin@gmail.com 3. The user enters the verification code 3.1. Example: 123457 4. The user clicks back on the popup message displaying login failure 5. The user enters the verification code 5.1. Example: 123456 6. The user navigates and clicks on the tour guide section 7. The user enters search details: example; 7.1. Destination: George Town 7.2. Duration: 13 August – 15 August 7.3. Attendees: 1 room for 2 adults and 1 child 8. The user clicks search button 9. The user clicks on Ahmad’s profile 10. The user clicks Book Now 11. The user clicks Proceed to Payment 12. The user enters their card details for payment 13. The user clicks the pay button 14. The user clicks the “Get booking confirmation”
Expected Test Outputs (with example)	1. A prompt appears requesting the user enter their email address

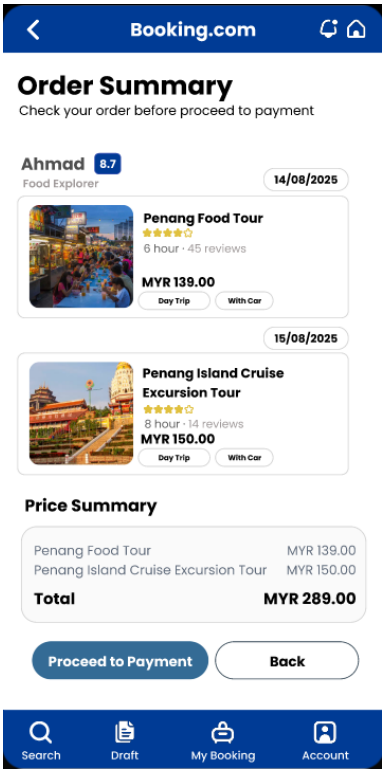
	2. A prompt appears to enter the verification code sent to the provided email 3. Login failure popup is displayed due to incorrect verification code 4. User is returned to the verification code entry screen 5. The user is successfully logged in and redirected to the main dashboard/homepage 6. A list or interface for tour guide services is displayed 7. Search form accepts input for destination, duration, and attendees 8. Search results show available tour guides based on search details 9. Ahmad's tour guide profile opens with details, pricing, availability, and reviews 10. Booking summary page appears with trip details, price, and a "Proceed to Payment" option 11. Payment page opens with fields for card/payment details 12. Card fields are populated correctly; system validates card format 13. Payment is processed. A success message 14. A confirmation page is displayed with booking reference number, details of the tour, and possibly an option to download or email the confirmation
Test Procedures	1. Enter "Continue with email" button 2. Enter your email address and click continue 3. Refer to the verification code sent by the system, enter the code and click verify email

	4. Click back on the Login failure pop-up message 5. Re-enter the correct verification code and click verify email 6. In the home page, navigate to the Tour Guide section using the header 7. Enter “George Town” for destination in the search tab 8. Enter 13 August to 15 August for duration in the search tab 9. Enter 2 adults 1 child for number of attendees in the search tab 10. Enter English for language preference on tour guide in your search tab 11. Click search 12. Browse through the list of tour guide 13. Click on Ahmad tour guide 14. Browse through Ahmad’s profile 15. Confirm itinerary by clicking “Book Now” button 16. Confirm total pricing and click “Proceed to Payment” 17. Select payment method “New card” 18. Enter your card details 19. Click “Pay MYR 289.00” for payment 20. Payment shows successfully booked the tour guide with payment details 21. Click on “Get boking confirmation” to proceed to confirmation page 22. Confirm that the tour guide booking displays on your booking page
Actual Test Results	

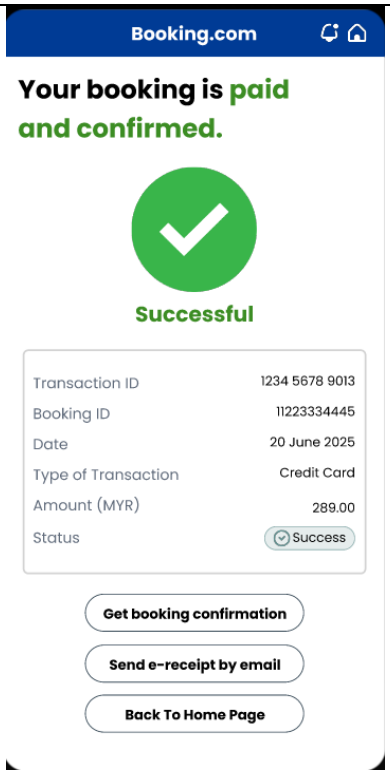
	1. Users are able to successfully login into their account after login failure: **successful**  2. User is able to search up list of tour guides: **successful**  3. User is able to view tour guides: **successful**



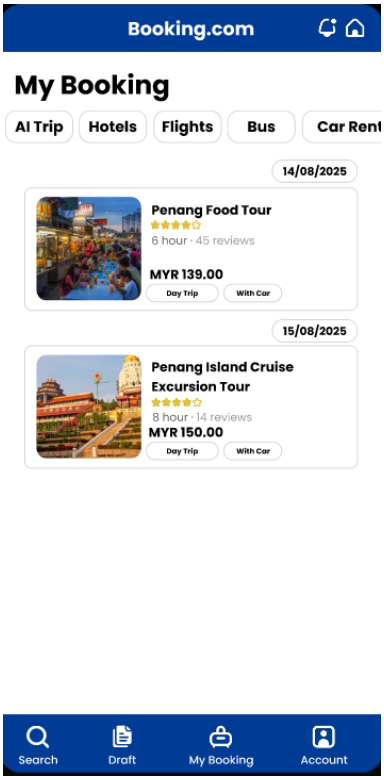
4. User is able to confirm booking with tour guide: **successful**



5. User is able to make payment: **successful**



6. User is able to view their booking confirmation: **successful**

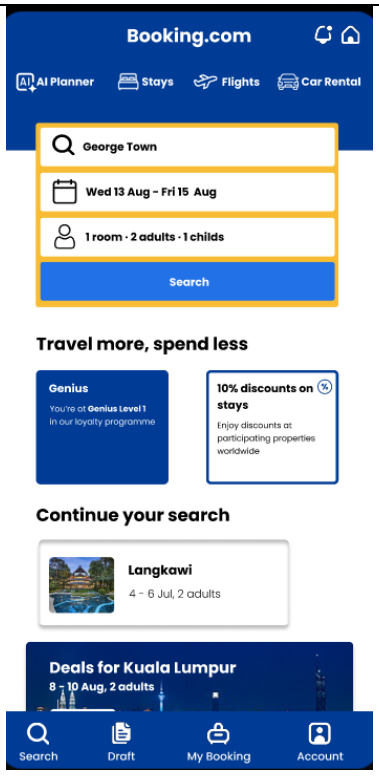


1.1.3 Test Case 3

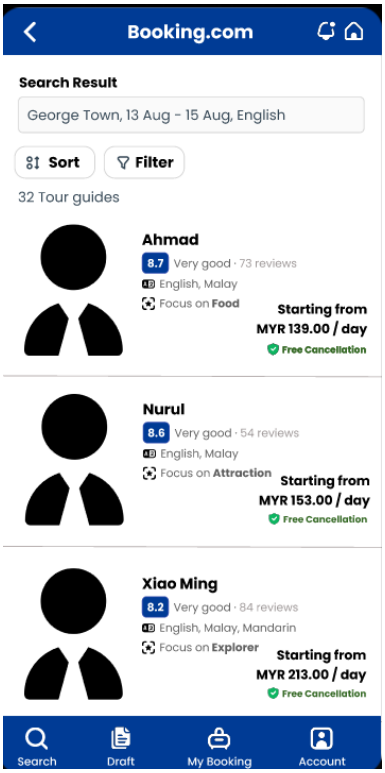
Test Objective	Alternate flow Explore other tour guide options when dissatisfied with the reviewed one during the testing of tour guide booking process
Potential Test Inputs (with example)	1. The user selects “Continue with email” as login method. 2. The user enters the email address. 2.1. Example: admin@gmail.com 3. The user enters the verification code 3.1. Example: 123456 4. The user navigates and clicks on the tour guide section 5. The user enters search details: example; 5.1. Destination: George Town 5.2. Duration: 13 August – 15 August 5.3. Attendees: 1 room for 2 adults and 1 child 6. The user clicks search button 7. The user clicks on Ahmad’s profile 8. The user clicks back to listing 9. The user selects another one 9.1. Example: Nurul 10. The user clicks Book Now 11. The user clicks Proceed to Payment 12. The user enters their card details for payment 13. The user clicks the pay button 14. The user clicks the “Get booking confirmation”
Expected Test Outputs (with example)	1. A prompt appears requesting the user enter their email address

	2. A prompt appears to enter the verification code sent to the provided email 3. The user is successfully logged in and redirected to the main dashboard/homepage 4. A list or interface for tour guide services is displayed 5. Search form accepts input for destination, duration, and attendees 6. Search results show available tour guides based on search details 7. Ahmad's tour guide profile opens with details, pricing, availability, and reviews 8. User is returned to the list of available tour guides 9. Nurul tour guide profile opens with details, pricing, availability, and reviews 10. Booking summary page appears with trip details, price, and a "Proceed to Payment" option 11. Payment page opens with fields for card/payment details 12. Card fields are populated correctly; system validates card format 13. Payment is processed. A success message 14. A confirmation page is displayed with booking reference number, details of the tour, and possibly an option to download or email the confirmation
Test Procedures	1. Enter "Continue with email" button 2. Enter your email address and click continue 3. Refer to the verification code sent by the system, enter the code and click verify email

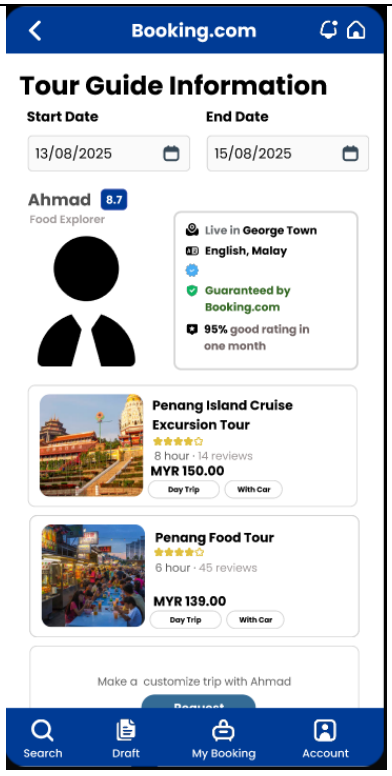
	4. In the home page, navigate to the Tour Guide section using the header 5. Enter “George Town” for destination in the search tab 6. Enter 13 August to 15 August for duration in the search tab 7. Enter 2 adults 1 child for number of attendees in the search tab 8. Enter English for language preference on tour guide in your search tab 9. Click search 10. Browse through the list of tour guide 11. Click on Ahmad tour guide 12. Browse through Ahmad’s profile 13. When unsatisfied, click “Back to Listing” 14. Click on Nurul 15. Browse through Nurul’s profile 16. Confirm itinerary by clicking “Book Now” button 17. Confirm total pricing and click “Proceed to Payment” 18. Select payment method “New card” 19. Enter your card details 20. Click “Pay MYR 238.00” for payment 21. Payment shows successfully booked the tour guide with payment details 22. Click on “Get boking confirmation” to proceed to confirmation page 23. Confirm that the tour guide booking displays on your booking page
Actual Test Results	<specify the actual outcome of the prototype> 1. Users are able to successfully login into their account: **successful**



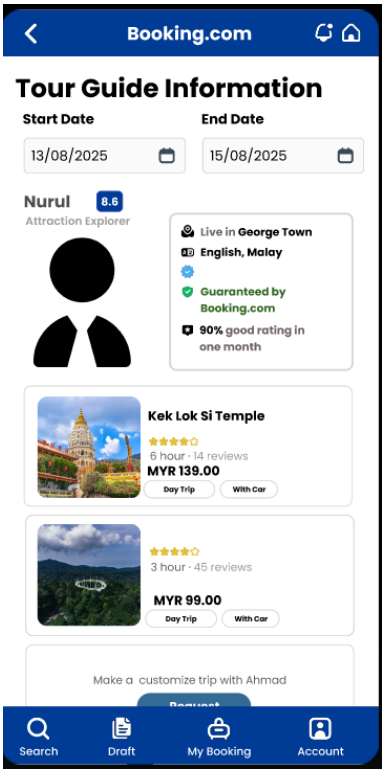
2. User is able to search up list of tour guides: **successful**



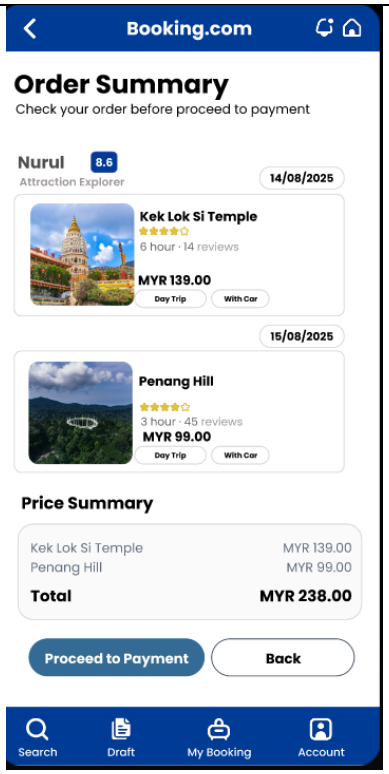
3. User is able to view tour guides: **successful**



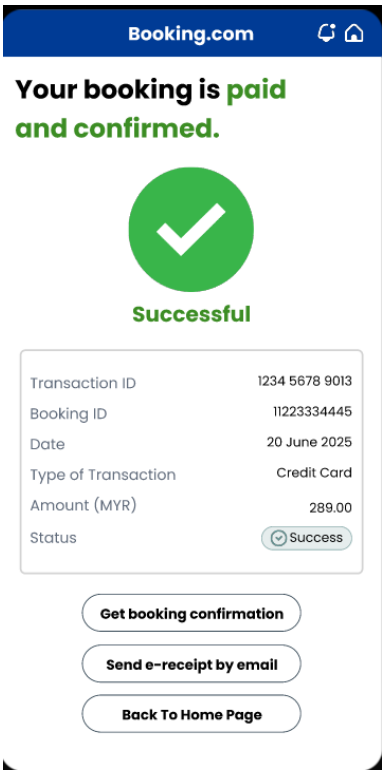
4. User is able to view other tour guides: **successful**



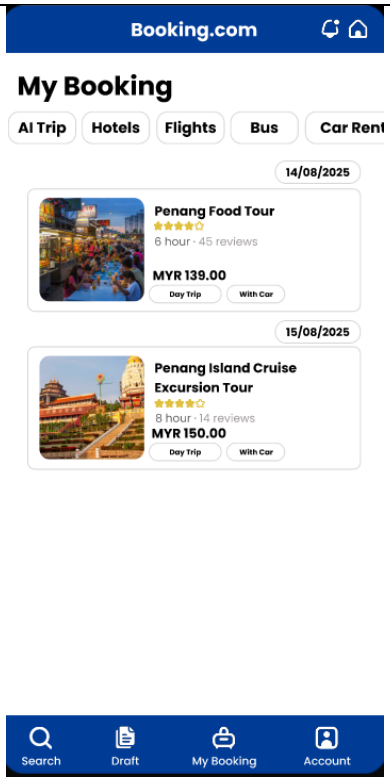
5. User is able to confirm booking with tour guide: **successful**



6. User is able to make payment: **successful**



7. User is able to view their booking confirmation: **successful**

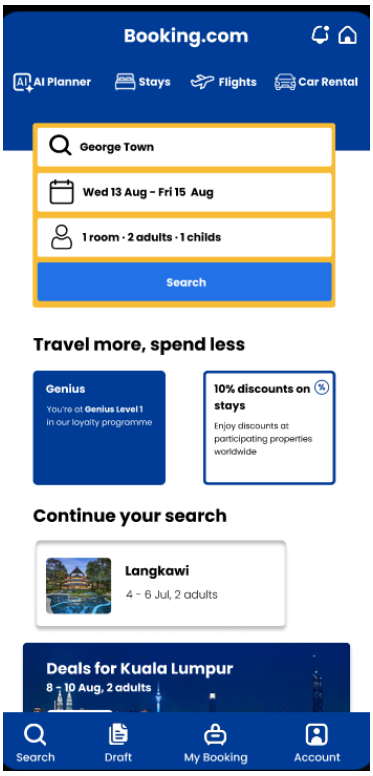
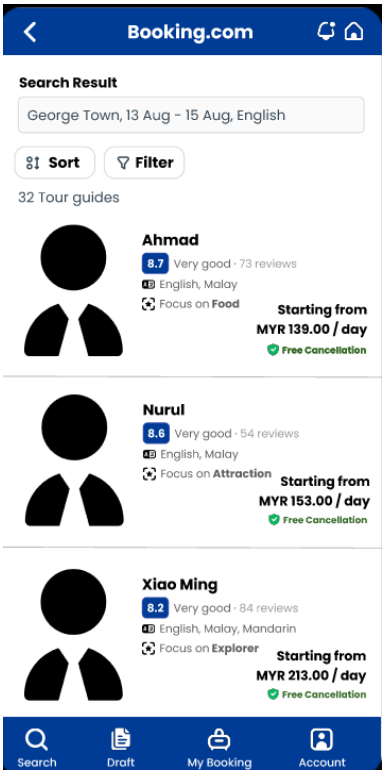
	 Booking.com My Booking AI Trip Hotels Flights Bus Car Rent 14/08/2025 Penang Food Tour 6 hour · 45 reviews MYR 139.00 Day Trip With Car 15/08/2025 Penang Island Cruise Excursion Tour 8 hour · 14 reviews MYR 150.00 Day Trip With Car Search Draft My Booking Account

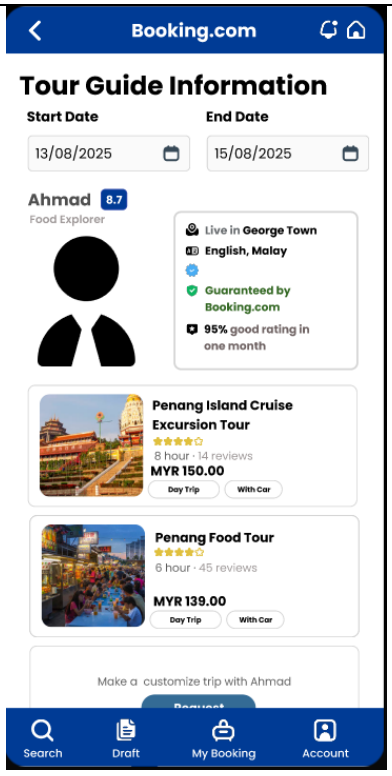
1.1.4 Test Case 4

Test Objective	Alternate flow Payment unsuccessful when testing the tour guide booking process
Potential Test Inputs (with example)	1. The user selects “Continue with email” as login method. 2. The user enters the email address. 2.1. Example: admin@gmail.com 3. The user enters the verification code 3.1. Example: 123456 4. The user navigates and clicks on the tour guide section 5. The user enters search details: example; 5.1. Destination: George Town 5.2. Duration: 13 August – 15 August 5.3. Attendees: 1 room for 2 adults and 1 child 6. The user clicks search button 7. The user clicks on Ahmad’s profile 8. The user clicks Book Now 9. The user clicks Proceed to Payment 10. The user enters their card details for payment 11. The user clicks the pay button 12. The user clicks back to go back to card details 13. The user enters their correct card details for payment 14. The user clicks the pay button 15. The user clicks the “Get booking confirmation”
Expected Test Outputs (with example)	1. A prompt appears requesting the user enter their email address

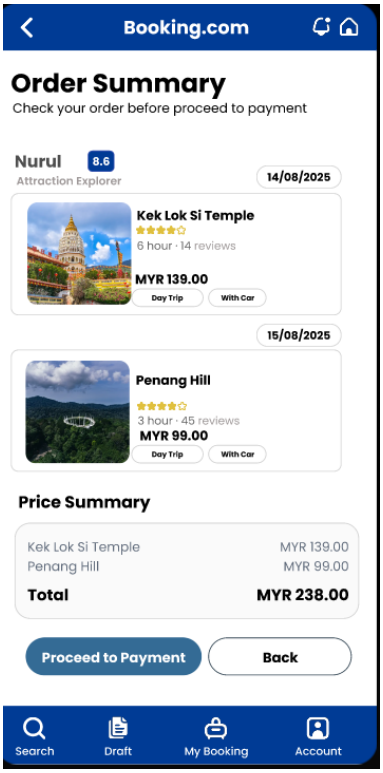
	2. A prompt appears to enter the verification code sent to the provided email 3. The user is successfully logged in and redirected to the main dashboard/homepage 4. A list or interface for tour guide services is displayed 5. Search form accepts input for destination, duration, and attendees 6. Search results show available tour guides based on search details 7. Ahmad's tour guide profile opens with details, pricing, availability, and reviews 8. Booking summary page appears with trip details, price, and a "Proceed to Payment" option 9. Payment page opens with fields for card/payment details 10. Card fields are entered; system validates card format 11. Payment is processed. An error message appears. 12. User is returned to the card details entry screen 13. Card fields are entered correctly; system validates card format 14. A confirmation page is displayed with booking reference number, details of the tour, and possibly an option to download or email the confirmation
Test Procedures	1. Enter "Continue with email" button 2. Enter your email address and click continue

	3. Refer to the verification code sent by the system, enter the code and click verify email 4. In the home page, navigate to the Tour Guide section using the header 5. Enter “George Town” for destination in the search tab 6. Enter 13 August to 15 August for duration in the search tab 7. Enter 2 adults 1 child for number of attendees in the search tab 8. Enter English for language preference on tour guide in your search tab 9. Click search 10. Browse through the list of tour guide 11. Click on Ahmad tour guide 12. Browse through Ahmad’s profile 13. Confirm itinerary by clicking “Book Now” button 14. Confirm total pricing and click “Proceed to Payment” 15. Select payment method “New card” 16. Enter your card details 17. Click “Pay MYR 289.00” for payment 18. Click “Pease try again” 19. Enter your card details correctly 20. Click “Pay MYR 289.00” for payment 21. Payment shows successfully booked the tour guide with payment details 22. Click on “Get boking confirmation” to proceed to confirmation page 23. Confirm that the tour guide booking displays on your booking page
Actual Test Results	<specify the actual outcome of the prototype>

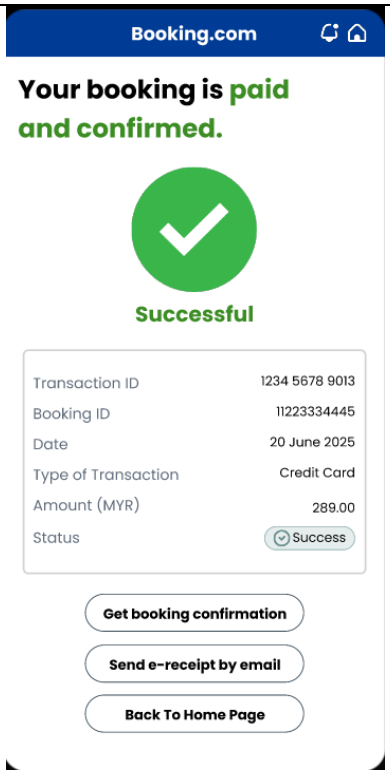
	1. Users are able to successfully login into their account: **successful**  2. User is able to search up list of tour guides: **successful**  3. User is able to view tour guides: **successful**



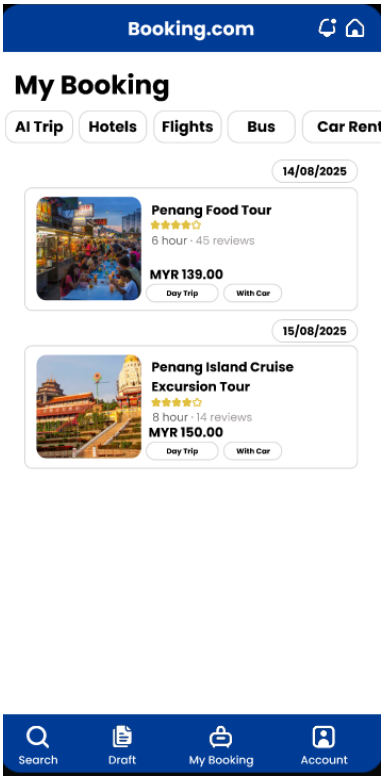
4. User is able to confirm booking with tour guide: **successful**



5. User is able to make payment after error: **successful**



6. User is able to view their booking confirmation: **successful**

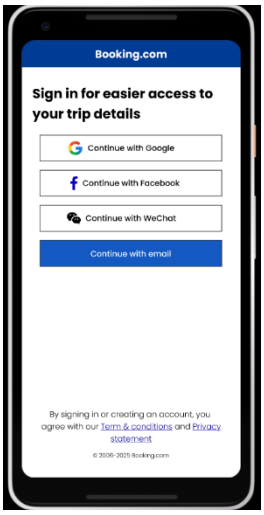
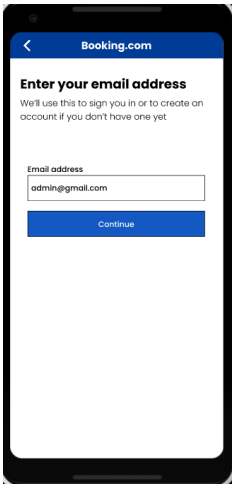


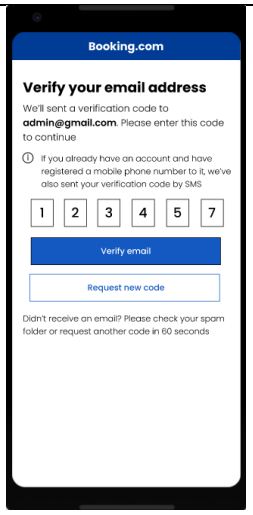
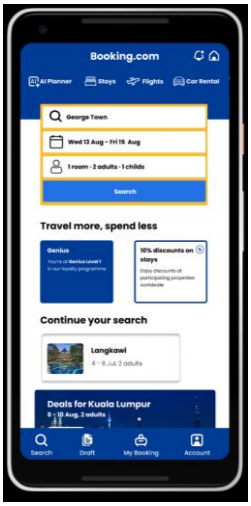
1.2 Manual Booking

Tester	Muhammad Hariz Hakimi Bin Mohd Hafiz
Test date	14th July 2024
Prototype developer	Liang Yrenn Da

1.2.1 Test Case 1

Test Objective	To test the login page
Potential Test Inputs (with example)	1. The user opens Booking.com Application. 2. The user selects “Continue with email” as login method. 3. The user enters the email address. 4. The user receives the verification code from email and fills in the code. 5. The user verifies the email to login. (**EX1**: Login failure, try to get the new code and verify again until successful) 6. The user is on the “home page”.
Expected Test Outputs (with example)	1. The user opens Booking.com Application. 2. The user selects “Continue with email” and fill their email. 3. The user receives the verification and fills in the code. 4. The user verifies the email to login. 5. The user is on the “home page”.
Test Procedures	1. The user opens Booking.com Application. 2. The user selects “Continue with email” as login method. 3. The user enters the email address, “admin@gmail.com”. 4. The user receives the verification code, “1 ,2 ,3 ,4 ,5 ,6” from email and fills in the code. 5. The user verifies the email to login. 6. The user is on the “home page”.

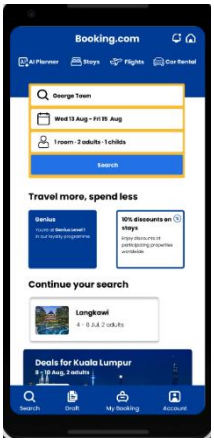
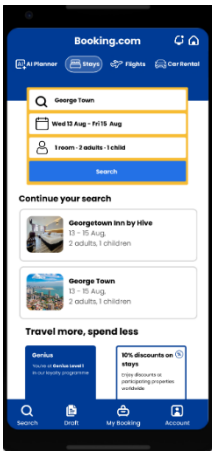
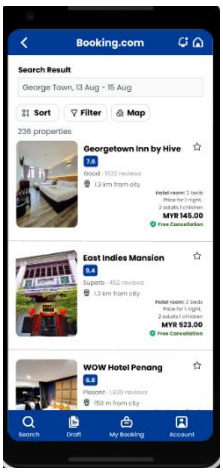
Actual Test Results	1. The user opens Booking.com Application: **Successful**  2. The user selects “Continue with email” and fill their email: **Successful**  3. The user receives the verification and fills in the code: **Successful**

	 4. The user verifies the email to login: **Successful** 5. The user is on the "home page": **Successful** 

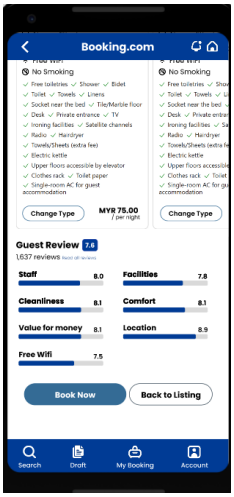
1.2.2 Test Case 2

Test Objective	To test the manual hotel booking flow process
Potential Test Inputs (with example)	1. The user opens the home page 2. The user clicks on "stays: to book hotel 3. The user searches for hotel by filling details: **Location**: George Town **Date**: 13 Aug – 15 Aug **Type**: 1 room – 2 adults – 1 child 4. The user clicks search to find suitable hotel 5. The user selects the hotel available

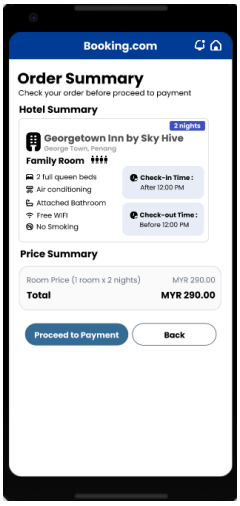
	6. The user will proceed to review the hotel summary 7. The user will proceed to payment window 8. The user will choose their payment method 9. The payment is successful 10. The user gets booking confirmation
Expected Test Outputs (with example)	1. The user opens the home page 2. The user clicks on “stays: to book hotel 3. The user searches for the hotel by filling in details: **Location**: George Town **Date**: 13 Aug – 15 Aug **Type**: 1 room – 2 adults – 1 child 4. The user clicks search to find suitable hotel 5. The user selects the hotel available 6. The user will proceed to review the hotel summary 7. The user will proceed to payment window 8. The user will choose their payment method 9. The payment is successful 10. The user gets to booking confirmation
Test Procedures	1. The user opens the home page 2. The user clicks on “stays: to book hotel 3. The user searches for the hotel by filling details: **Location**: George Town **Date**: 13 Aug – 15 Aug **Type**: 1 room – 2 adults – 1 child 4. The user clicks search to find suitable hotel 5. The user selects the hotel available 6. The user will proceed to review the hotel summary 7. The user will proceed to payment window

	The user will choose their payment method 8. The payment is successful 9. The user gets booking confirmation
Actual Test Results	1. The user opens the homepage: **successful**  2. The user selects “stays” to book hotel: **successful**  3. The user clicks search to review hotel available**: successful** 

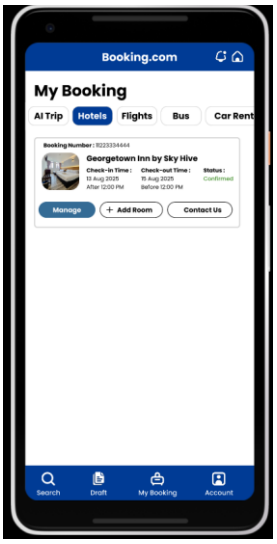
4. The user selects book now: **successful**



5. The user proceed payment: **successful**

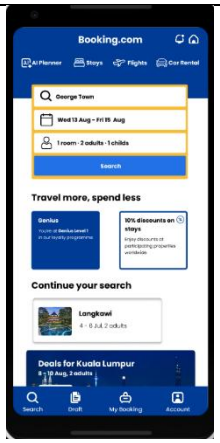


6. The user receives booking information: **successful**

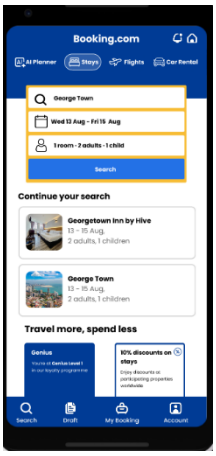


1.2.3 Test Case 3

Test Objective	To test the flow of booking from past searches
Potential Test Inputs (with example)	1. The user opens the home page 2. The user selects “stay” to proceed to hotel booking 3. The user selects the hotel from past searches 4. The user will proceed to review the hotel summary 5. The user will proceed to payment window 6. The user will choose their payment method 7. The payment is successful 8. The user gets to booking confirmation
Expected Test Outputs (with example)	1. The user opens the home page 2. The user selects “stay” to proceed to hotel booking 3. The user selects the hotel from past searches 4. The user will proceed to review the hotel summary 5. The user will proceed to payment window 6. The user will choose their payment method 7. The payment is successful 8. The user gets booking confirmation
Test Procedures	1. The user opens the home page 2. The user selects “stay” to proceed to hotel booking 3. The user selects the hotel from past searches 4. The user will proceed to review the hotel summary 5. The user will proceed to payment window 6. The user will choose their payment method 7. The payment is successful 8. The user gets booking confirmation
Actual Test Results	1. The user opens the homepage: **successful**



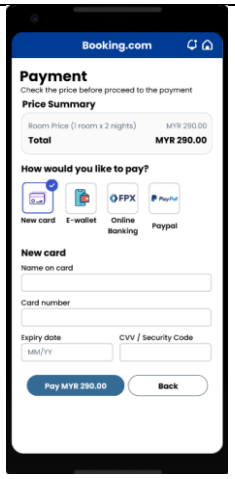
2. The user selects “stays” to book hotel: **successful**



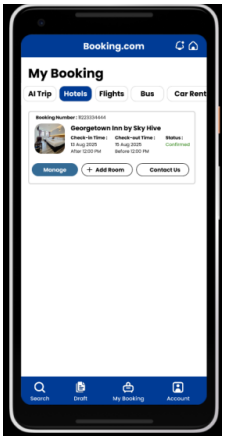
3. The user clicks on past hotel searches to book**:** **successful**
4. The user selects book now: **successful**



5. The user proceed payment: **successful**



6. The user receives booking information: **successful**

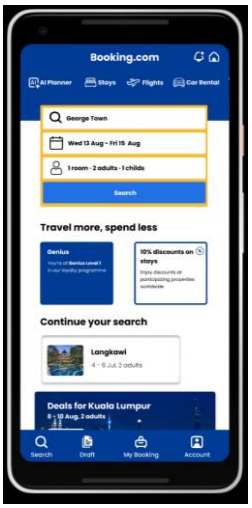


1.3 Transportation Booking

Tester	Yeoh Jie Long
Test date	12th July 2024
Prototype developer	Ooi Jin Yon

1.3.1 Test Case 1

Test Objective	To test the login page
Potential Test Inputs (with example)	1. The user opens Booking.com Application. 2. The user selects “Continue with email” as login method. 3. The user enters the email address. 4. The user receives the verification code from email and fills in the code. 5. The user verifies the email to login. (**EX1**: Login failure, try to get the new code and verify again until successful) 6. The user is on the “home page”.
Expected Test Outputs (with example)	1. The user opens Booking.com Application. 2. The user selects “Continue with email” and fill their email. 3. The user receives the verification and fills in the code. 4. The user verifies the email to login. 5. The user is on the “home page”.
Test Procedures	1. The user opens Booking.com Application. 2. The user selects “Continue with email” as login method. 3. The user enters the email address, “admin@gmail.com”. 4. The user receives the verification code, “1 ,2 ,3 ,4 ,5 ,6” from email and fills in the code.

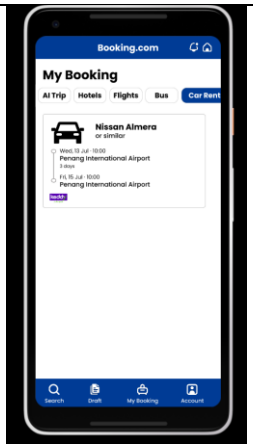
	5. The user verifies the email to login. 6. The user is on the “home page”.
Actual Test Results	1. The user opens Booking.com Application: **Successful** 2. The user selects “Continue with email” and fill their email: **Successful** 3. The user receives the verification and fills in the code: **Successful** 4. The user verifies the email to login: **Successful** 5. The user is on the “home page”: **Successful** 

1.3.2 Test Case 2

Test Objective	To test the car rental booking flow
Potential Test Inputs (with example)	1. The user is on “Car Rental” 2. They search for buses by filling in the details: **Pick up location:** “Penang International Airport” **Duration:** “13 Aug – 15 Aug” **Driver’s age:** “20-65”. 3. They click on “Search” button. 4. They browse and select a suitable rental car. 5. They proceed to the specific rental car information page.

	6. After they view the full information, they will proceed to payment. 7. Before payment, they can choose to add protection or not. 8. They will look through the booking summary 9. They will proceed with the payment process 10. They choose their payment method 11. Their payment is successful. (**EX1**: The payment may be unsuccessful, and need make the payment again until successful) 12. Back to booking confirmation page.
Expected Test Outputs (with example)	1. The user is on “Car Rental” 2. They are searching for a rental car that fits their requirements. 3. The available car rental option has been listing out for them to choose the most suitable. **Duration:** 13 Aug – 15 Aug **Pick-up location:** Penang International Airport 4. A few suitable car rental options will be shown, and the user will choose the most suitable from below: **Duration:** 13 Aug – 15 Aug **Type of Car:** Nissan Almera, Proton Persona, Honda HR-V, Nissan X-Trail. 5. The user selects the car and proceeds to payment 6. The user selects to add on the travel protection or not. 7. The user selects a suitable payment method: New Card, E-wallet, Online banking, PayPal 8. They successfully book their rental car on 13 Aug – 15 Aug in Penang International Airport.
Test Procedures	1. The user is on “Car Rental”

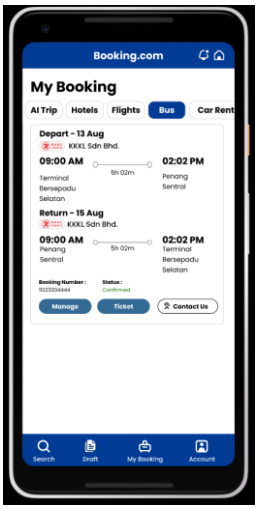
	2. They search for buses by filling in the details: **Pick up location:** “Penang International Airport” **Duration:** “13 Aug – 15 Aug” **Driver’s age:** “20-65”. 3. They click on “Search” button. 4. They browse and select a suitable rental car. 5. They proceed to the specific rental car information page. 6. After they view the full information, they will proceed to payment. 7. Before payment, they can choose to add protection or not. 8. They look through the booking summary 9. They proceed with the payment process 10. They choose “New Card” as their payment method 11. Their payment is successful. 12. Back to booking confirmation page.
Actual Test Results	1. The user in on “Car Rental”: **Successful** 2. They selected a suitable rental car based on the requirements: **Successful** 3. They selected the addon travel protection insurance: **Successful** 4. They select their payment method and succeed in making their payment: **Successful** 5. They go back to homepage or redirect to booking confirmation page: **Successful**

1.3.3 Test Case 3

Test Objective	To test the bus page booking flow.
Potential Test Inputs (with example)	1. The user is on “Bus Ticket” 2. They chose their bus option, either round trip or one way. 3. They search for buses by filling in the details. **Departure**: “Terminal Bersepadu Selatan” **Arrival**: “Penang Sentral (Butterworth)” **Number of persons**: “2 adults, 1 child” 4. They browse and select a suitable depart bus ticket. 5. They will choose their seat 6. They browse and select a suitable return bus ticket. 7. They will choose their seat 8. They will view their order summary 9. After they view the summary, they will go to the payment page. 10. They choose their payment method 11. Their payment is successful. (**EX1**: The payment may be unsuccessful, and need make the payment again until successful) 12. Back to booking confirmation page.

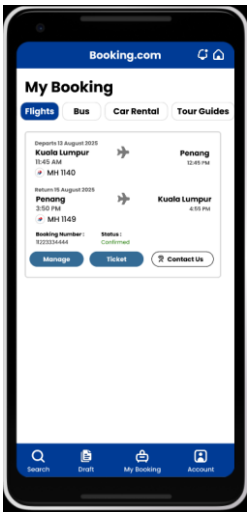
Expected Test Outputs (with example)	1. The user is on “Bus ticket” 2. They are searching for a bus ticket that fits their requirements. 3. The available bus ticket has been listing out for them to choose the most suitable. **Departure**: “Terminal Bersepadu Selatan” **Arrival**: “Penang Sentral (Butterworth)” **Number of persons**: “2 adults, 1 child” 4. They click on “Search” button. 5. The user selects the bus and picks their seat for a two-way ticket. 6. They proceed to payment page 7. The user selects a suitable payment method: New Card, E-wallet, Online banking, PayPal 8. They successfully booked their two-way bus ticket from TBS to Penang Sentral.
Test Procedures	1. The user is on “Bus Ticket” 2. They chose their bus option as a round trip 3. They search for buses by filling in the details. **Departure**: “Terminal Bersepadu Selatan” **Arrival**: “Penang Sentral (Butterworth)” **Number of persons**: “2 adults, 1 child” 4. They select a suitable depart bus ticket. 5. They choose their seat 6. They select a suitable return bus ticket. 7. They choose their seat 8. They will view their order summary 9. They will go to the payment page. 10. They choose “new card” as their payment method. 11. Their payment is successful. 12. Back to homepage or booking confirmation page.
Actual Test Results	1. The user is on “Bus ticket”: **Successful**

	2. They selected a suitable bus ticket based on the requirements: **Successful** 3. They selected their seat: **Successful** 4. They selected their payment method and succeeded in making their payment: **Successful** 5. They go back to homepage or redirect to booking confirmation page: **Successful** 

1.3.4 Test Case 4

Test Objective	To test the flight page booking flow.
Potential Test Inputs (with example)	1. The user is on “Flight” 2. They chose their flight option, either round trip or one way. 3. They search for flight by filling in the details. **Departure**: “Kuala Lumpur – All Airports” **Arrival**: “Penang International Airport” **Number of persons**: “2 adults, 1 child” 4. They browse and select a suitable flight. 5. They will choose their desired time. 6. They browse and select a suitable return flight. 7. They will choose their desired time 8. They will view their order summary

	9. After they view the summary, they will go to the payment page. 10. They choose their payment method 11. Their payment is successful. (**EX1**: The payment may be unsuccessful, and need make the payment again until successful) 12. Back to booking confirmation page.
Expected Test Outputs (with example)	1. The user is on “Flight” 2. They are searching for a flight that fits their requirements. 3. The available flight has been listing out for them to choose the most suitable. **Departure**: “Kuala Lumpur – All Airports” **Arrival**: “Penang International Airport” **Number of persons**: “2 adults, 1 child” 4. They click on “Search” button. 5. The user selects the flight with the desired time for a two-way ticket. 6. They proceed to payment page 7. The user selects a suitable payment method: New Card, E-wallet, Online banking, PayPal 8. They successfully booked their two-way flight ticket from Kuala Lumpur – All Airports to Penang International Airport.
Test Procedures	1. The user is on “Flight” 2. They chose their flight option as a round trip 3. They search for buses by filling in the details. **Departure**: “Kuala Lumpur – All Airports” **Arrival**: “Penang International Airport” **Number of persons**: “2 adults, 1 child” 4. They select a suitable depart flight. 5. They choose their desired time 6. They select a suitable return flight.

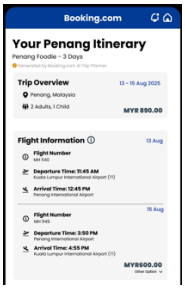
	7. They choose their desired time 8. They will view their order summary 9. They will go to the payment page. 10. They choose “new card” as their payment method. 11. Their payment is successful. 12. Back to homepage or booking confirmation page.
Actual Test Results	1. The user is on “Flight”: **Successful** 2. They selected a suitable flight based on the requirements: **Successful** 3. They selected their desired time: **Successful** 4. They selected their payment method and succeeded in making their payment: **Successful** 5. They go back to homepage or redirect to booking confirmation page: **Successful** 

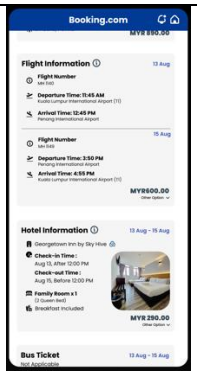
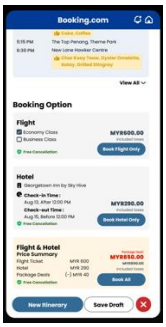
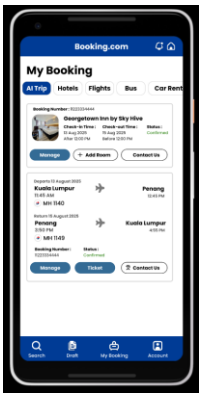
1.4 AI Trip Planner

Tester	Wong Xu Liang
Test date	12 July 2025
Prototype developer	Ooi Jin Yon, Liang Yrenn Da, Yeoh Jie Long

1.4.1 Test Case 1

Test Objective	To test AI trip planner function with present use of full booking
Potential Test Inputs (with example)	1. User accesses the “AI Trip Planner” module. 2. Trip details are input: • From: Kuala Lumpur • To: George Town • Date: 13/8/2025 to 15/8/2025 • Budget: RM1000 • Group: 2 Adults, 1 Child • Travel Style: Foodie • Special Notes: "Play around in George Town and try Penang food" 3. User clicks “Start Generate Itinerary”.
Expected Test Outputs (with example)	1.Itinerary is successfully generated. 2.It includes: • Flight info: e.g., Flight AK6123, RM320 total • Hotel info: e.g., Evergreen Laurel Hotel, RM420 total • Trip itinerary for Day 1 and Day 2 with food stops and activities • Booking options shown: [Book Flight Only], [Book Food Only], [Book All] 3.User selects “Book All”. 4.Display of: • Flight summary

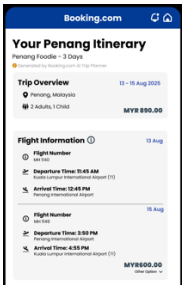
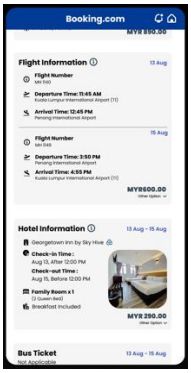
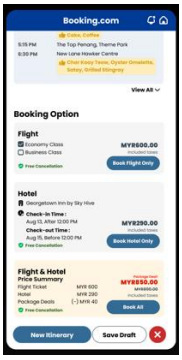
	• Hotel summary • Price summary (RM980) 5. User proceeds to payment and completes transaction. 6.Redirected to Booking Confirmation Page.
Test Procedures	1. Navigate to AI Trip Planner module. 2. Input: • KL to George Town, 13–15 Aug 2025 • RM1000 budget • 2 adults, 1 child • Style: Foodie • Special Note: “Play around in George Town and try Penang food” 3. Click “Start Generate Itinerary”. 4. Review generated itinerary (flights, hotel, day plans). 5. Click “Book All”. 6. Review full booking summary. 7. Choose payment method (e.g., Online Banking). 8. Complete payment. 9. Confirm redirection to Booking Confirmation Page.
Actual Test Results	1. Inputs accepted and itinerary generated: **Successful**  2. Flight, hotel, and itinerary details accurate: **Successful**

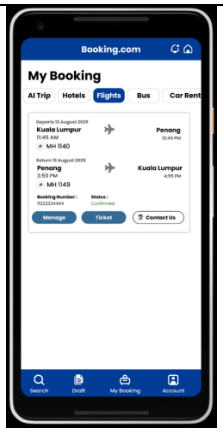
	 3. Flight and Hotel Booking “Book All” function worked and showed full summary: **Successful**  4. Payment completed and confirmation displayed: **Successful** 

1.4.2 Test Case 2

Test Objective	To test the AI Trip Planner module with “Flight Only” booking option.
Potential Test Inputs (with example)	1) User accesses the “AI Trip Planner” module. 2) Trip details are input: • From: Kuala Lumpur • To: George Town

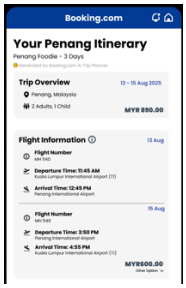
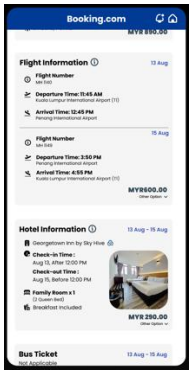
	• Date: 13/8/2025 to 15/8/2025 • Budget: RM1000 • Group: 2 Adults, 1 Child • Travel Style: Foodie • Special Notes: "Play around in George Town and try Penang food" 3) User clicks “Start Generate Itinerary”.
Expected Test Outputs (with example)	1. Itinerary is successfully generated. 2. It includes: • Flight info: e.g., Firefly FY1004, RM300 total • Hotel info (shown but not booked) 3. Trip itinerary with food spots Booking options: [Book Flight Only], [Book Food Only], [Book All] 4. User selects “Book Flight Only”. 5. Display of: • Flight summary only • Total price: RM300 6. User proceeds to payment. 7. Confirmation page displays only flight booking.
Test Procedures	1. Access AI Trip Planner module. 2. Input travel details: • KL to George Town, 13–15 Aug 2025 • Budget RM1000 • 2 adults, 1 child • Style: Foodie • Notes: “Play around in George Town and try Penang food” 3. Generate itinerary. 4. Select “Book Flight Only”. 5. Review flight details. 6. Proceed with payment via E-wallet.

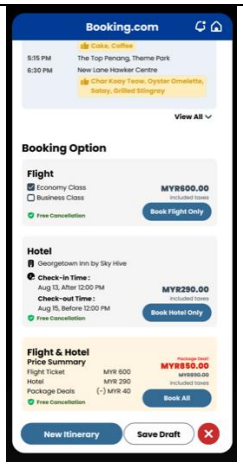
	7. Confirm successful redirection to confirmation page.
Actual Test Results	1. Itinerary generated: **Successful**  2. Flight details accurate: **Successful**  3. “Book Flight Only” option showed correct summary: **Successful**  4. Payment processed and booking confirmed: **Successful**

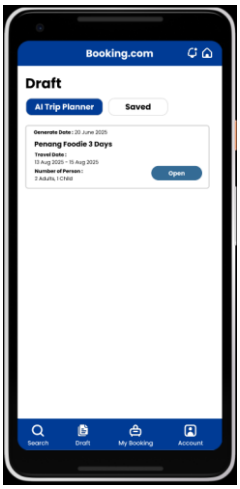
1.4.3 Test Case 3

Test Objective	To test the AI Trip Planner module with “Flight Only” booking option.
Potential Test Inputs (with example)	1) User accesses the “AI Trip Planner” module. 2) Trip details are input: • From: Kuala Lumpur • To: George Town • Date: 13/8/2025 to 15/8/2025 • Budget: RM1000 • Group: 2 Adults, 1 Child • Travel Style: Foodie • Special Notes: "Play around in George Town and try Penang food" 3) User clicks “Start Generate Itinerary”.
Expected Test Outputs (with example)	1. Itinerary is successfully generated. 2. It includes: • Flight info: e.g., Firefly FY1004, RM300 total • Hotel info (shown but not booked) 3. Trip itinerary with food spots Booking options: [Book Flight Only], [Book Food Only], [Book All] 4. User selects “Save Draft”.

	5. Display of: • Penang Foodie 3 Days • Travel Date • Number of Person
Test Procedures	1. Access AI Trip Planner module. 2. Input travel details: • KL to George Town, 13–15 Aug 2025 • Budget RM1000 • 2 adults, 1 child • Style: Foodie • Notes: “Play around in George Town and try Penang food” 3. Generate itinerary. 4. Select “Save Draft”. 5. Display of AI draft saved
Actual Test Results	1. Itinerary generated: **Successful**  2. Flight & Hotel details accurate: **Successful**  3. “Save Draft” option showed correct summary: **Successful**



4. Display of AI draft saved: **Successful**



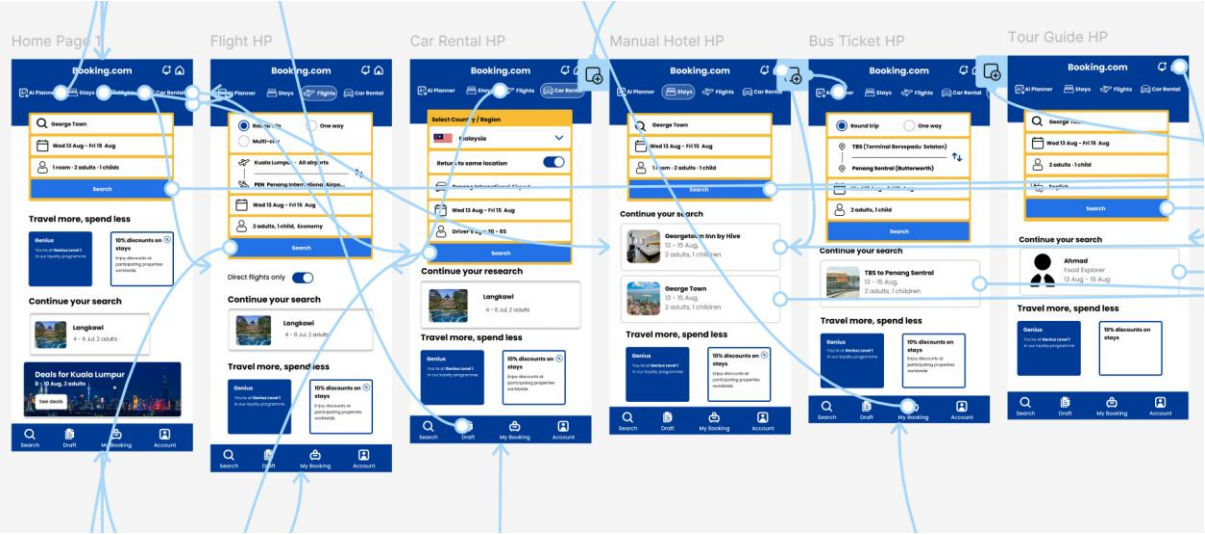
2.0 Initial Prototype

Login Page



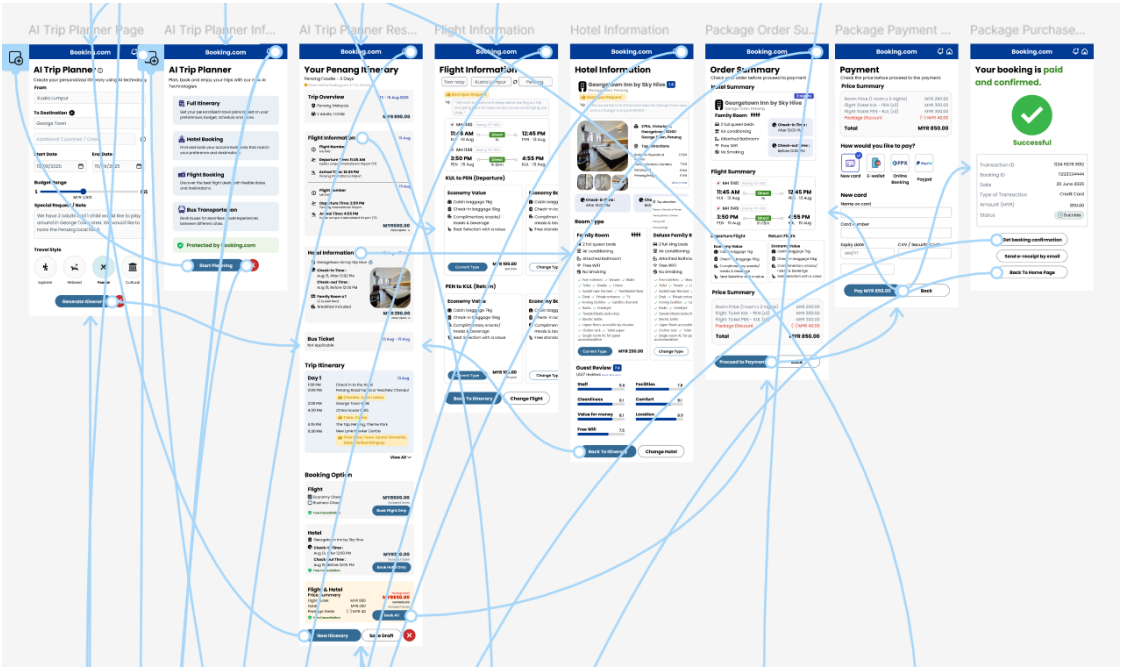
Before user access to Booking.com application, the user are required to login their account using Google, Facebook, WeChat or an email. From this scenario, users will use email to login. They will receive the verification code to verify whether they use their own account. If the verification code is incorrect, they will need to verify again with the correct code so they can login to the home page.

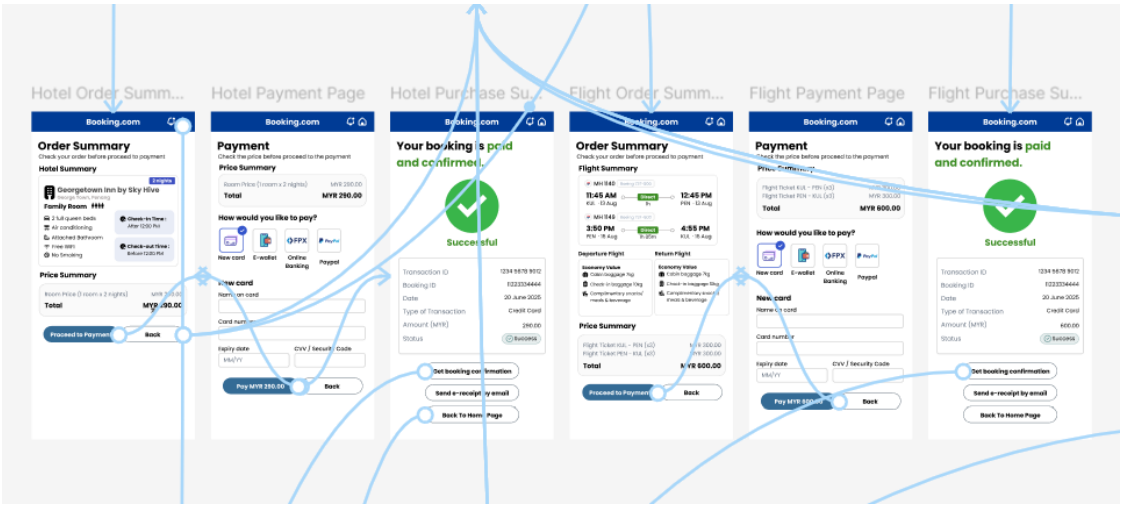
Home Page



After user login, they will redirect to the home page. From homepage, they can access to AI trip planner, flight homepage, hotel homepage, tour guide homepage, bus ticket homepage and car rental homepage. They can choose which section they want to browse to find the thing they want.

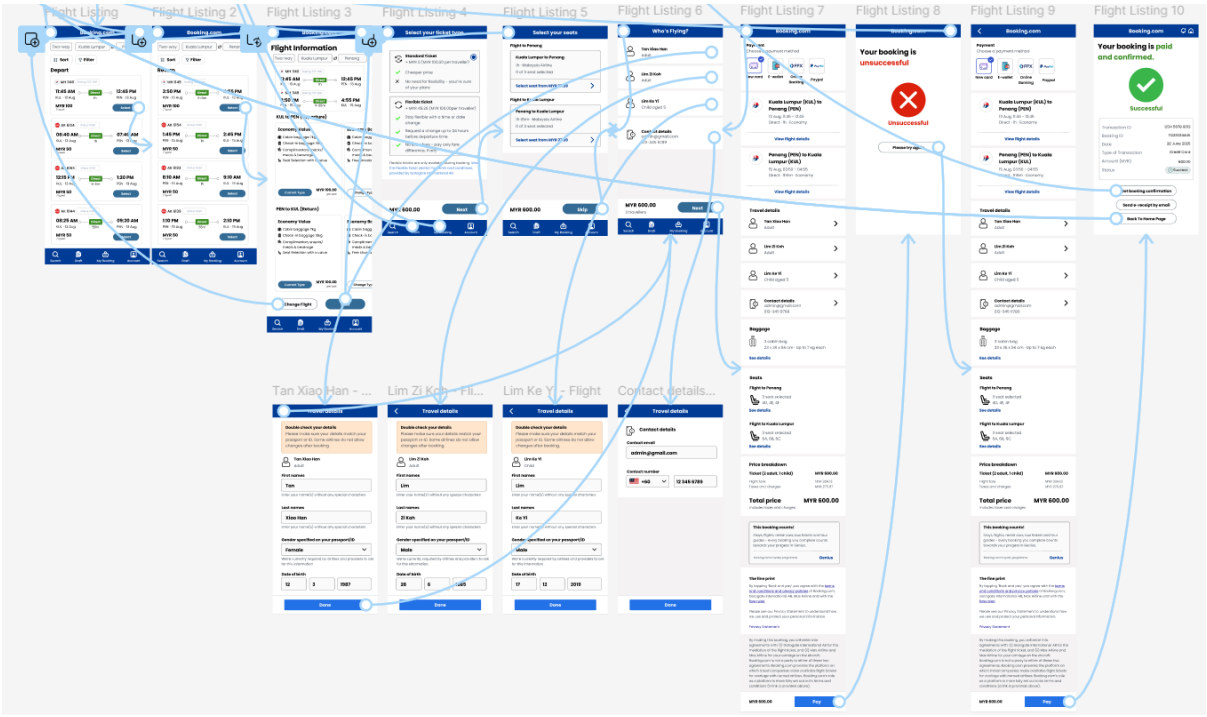
AI Trip Planner





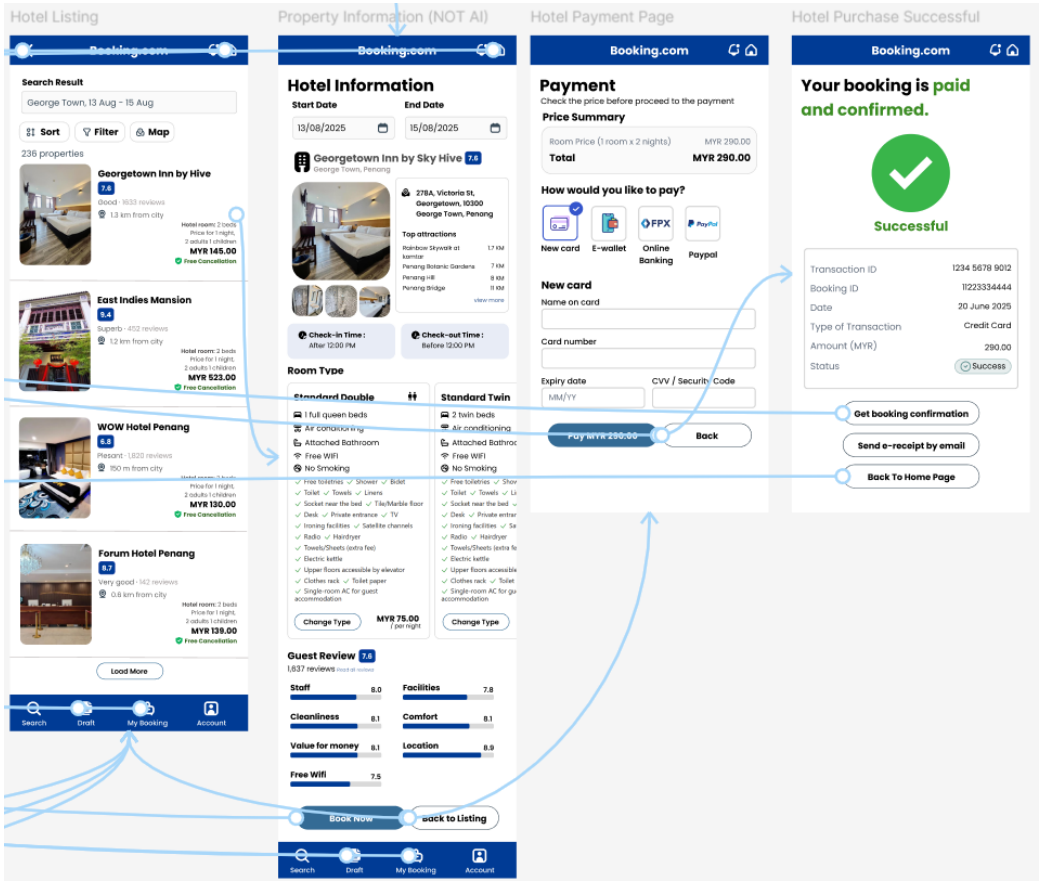
This is the page after the user click on AI trip planner button from homepage. The AI Trip planner started with the user input and request to generate an itinerary that matches their requests. After they key in the destination, number of persons, duration, budget and other request, the AI Trip planner will start to generate itinerary with flight, hotel, bus and tour guide (upon request) option for them to book. They can choose to buy one of them or buy as a package. When they click into the button, they will go for their detailed information to ensure it fulfil their satisfaction. Once they confirmed everything, they will proceed to payment. After the payment successful, they will get their booking confirmation.

Flight Booking Flow



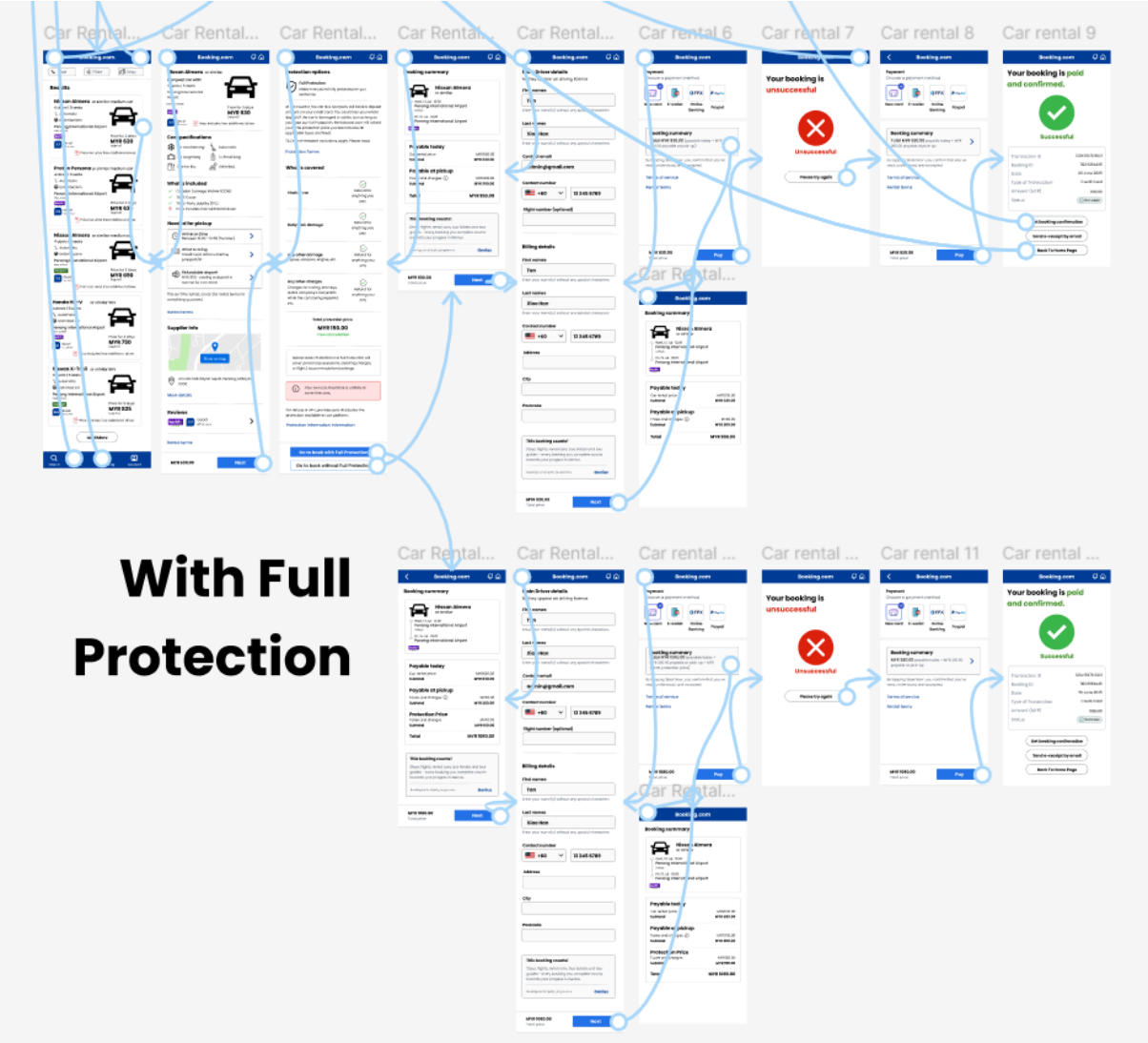
This is the result after they click the “Search” button from flight home page. As they choose round trip from Kuala Lumpur to Penang during 13 August to 15 August, a list of suitable flight is shown for the user to select their departure and return ticket. After that, they will be asked whether they want to remain as standard ticket or upgrade to flexible ticket that allowed them to change ticket easily. After they select, they will be able to select their seats and are required to put in passenger information with one contact detail so the airline company able to send any changes on their flight. Finally, they will make the payment and receive their booking confirmation.

Hotel Booking Flow



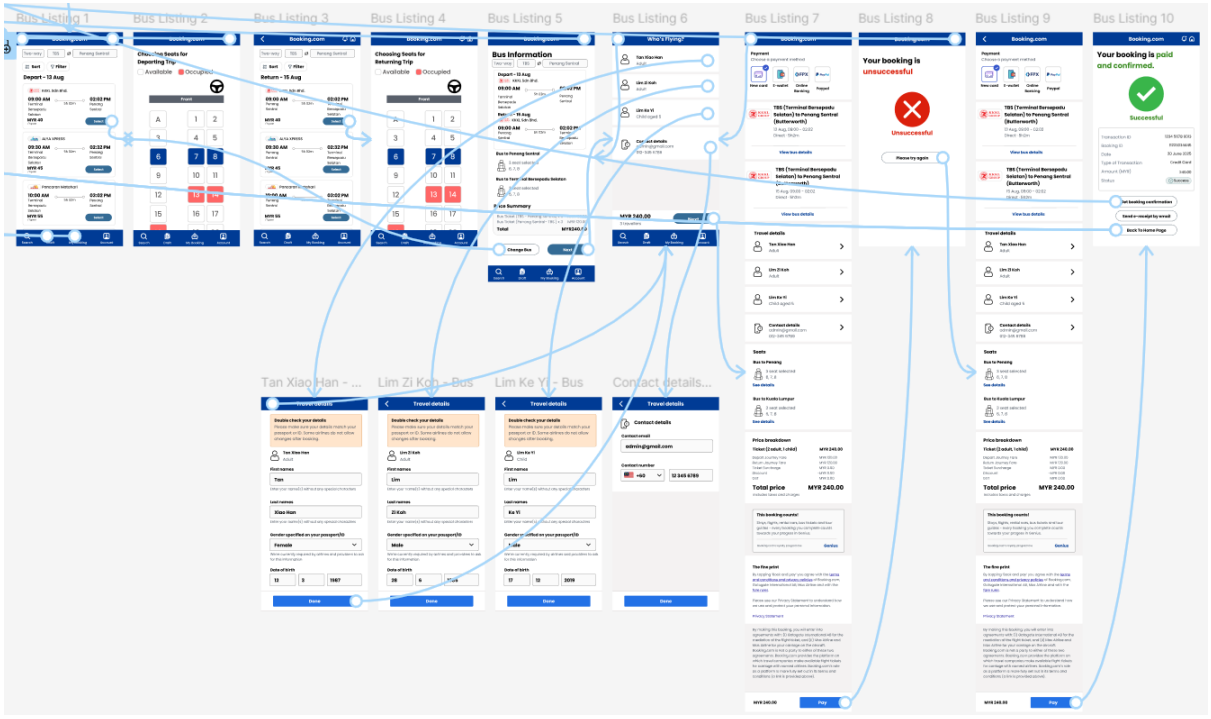
This is the result after they click the “Search” button from hotel home page. The user chooses their destination, duration, number of peoples and number of rooms. The system will list out the suitable hotel results. The user will choose the one they like and see more information when they click in. If they are satisfying with the hotel, they will click on “Book Now” to the payment page. After payment successful, they will receive their booking confirmation.

Car Rental Flow



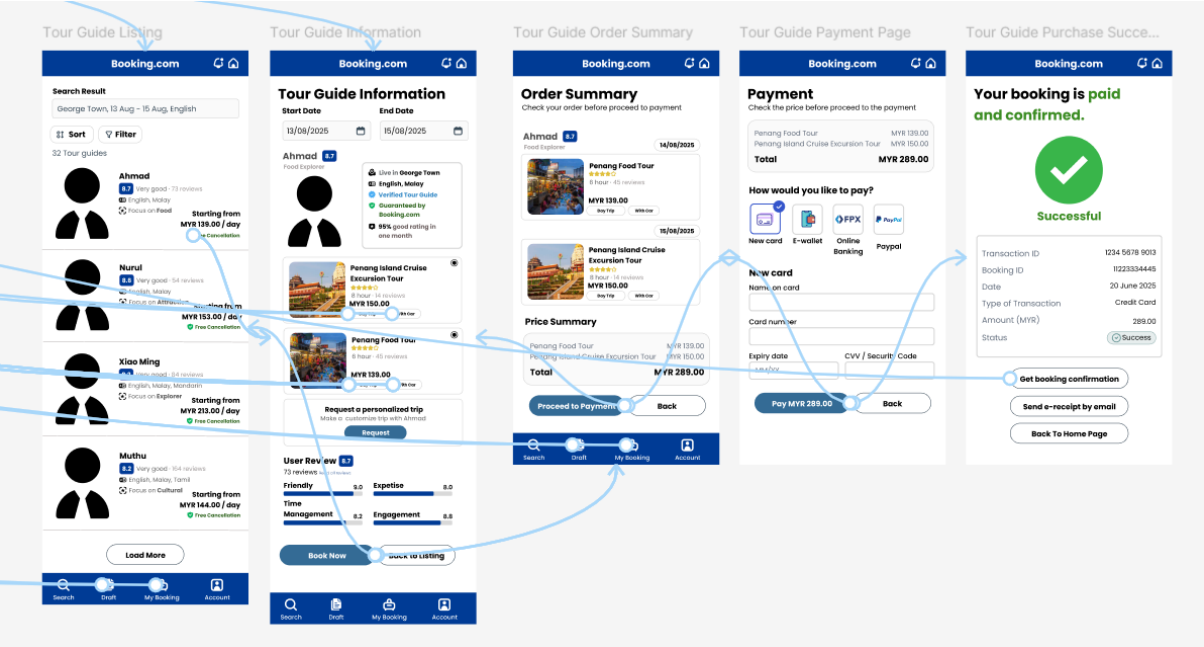
This is the result after they click the “Search” button from car rental home page. The user chooses their pickup location, duration, number of peoples and driver age. The system will list out the suitable rental car results. The user will choose the one they like and see more information when they click in. If they are satisfying with the rental car selection, they will proceed to next step to decide whether they need travel protection or not. After that, they will proceed to the payment page. Finally, the booking is done, and they will receive their booking confirmation.

Bus Ticket Flow



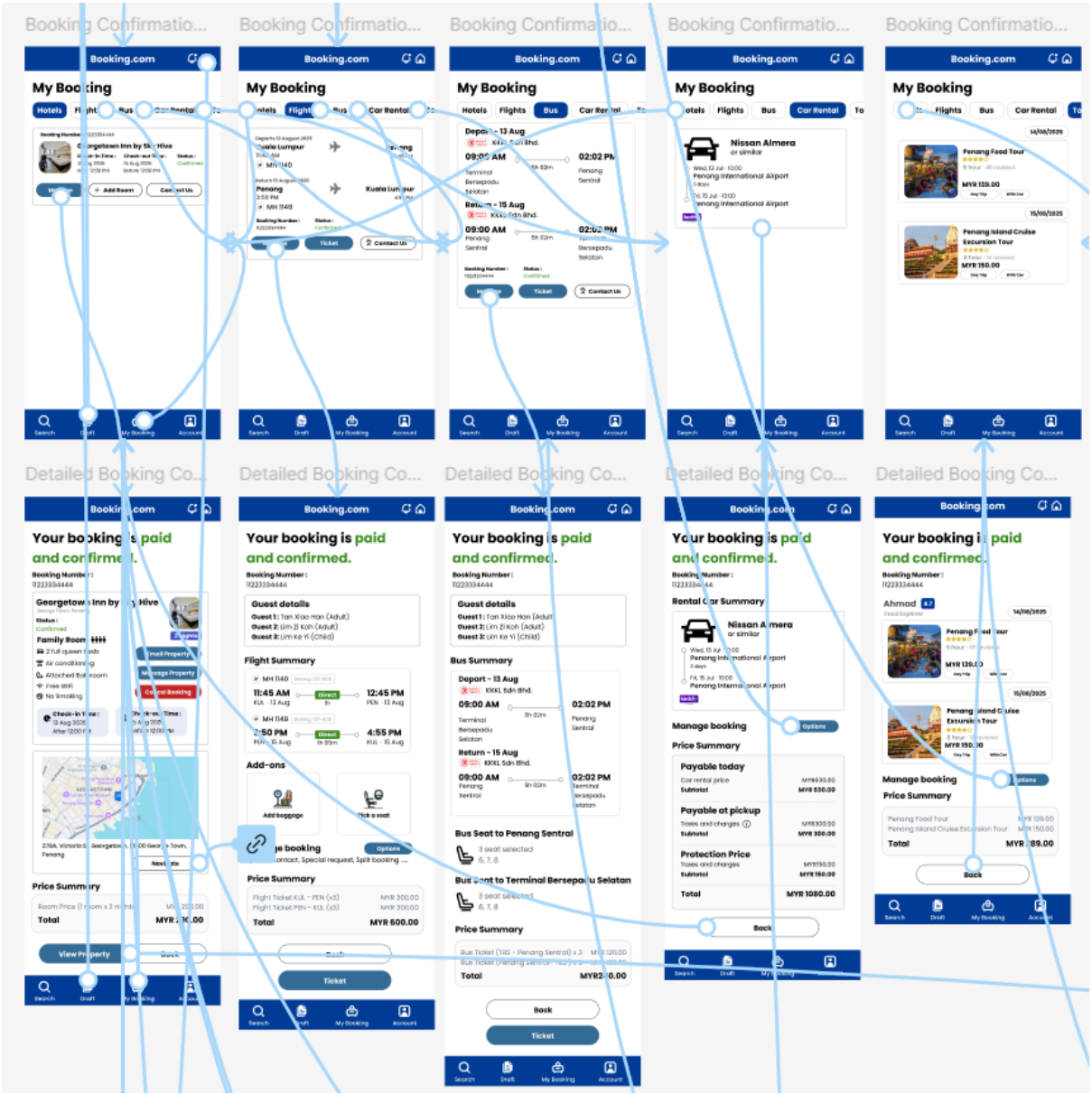
This is the result after they click the “Search” button from bus ticket home page. As they choose round trip from Terminal Bersepadu Selatan to Penang Sentral during 13 August to 15 August, a list of suitable bus is shown for the user to select their departure and return ticket. They will be able to select their seats and are required to put in passenger information with one contact detail so the bus company able to send any changes on their bus. Finally, they will make the payment and receive their booking confirmation.

Tour Guide Flow



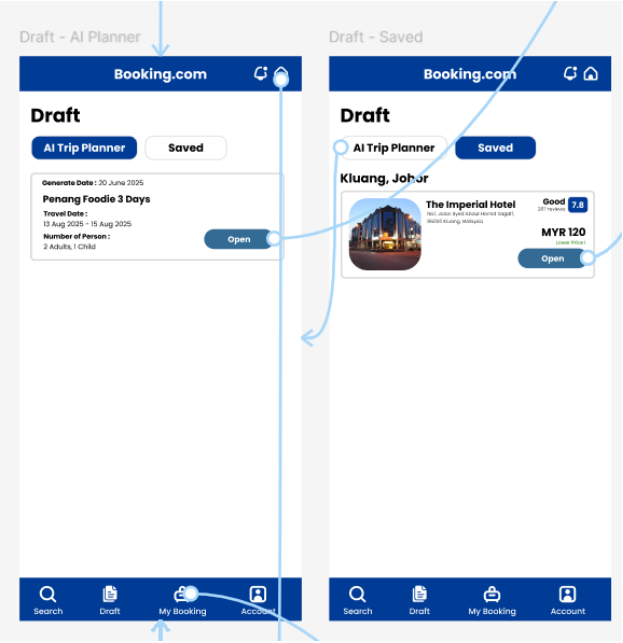
This is the result after they click the “Search” button from tour guide home page. The user chooses their destination, duration, number of peoples and preferred language. The system will list out the suitable tour guide results. The user will choose their favour tour guide and see more information when they click in. If they are satisfying with the tour provided from them, they will proceed to the payment page. If the payment successful, the booking is done, and they will receive their booking confirmation.

Booking Confirmation Page`



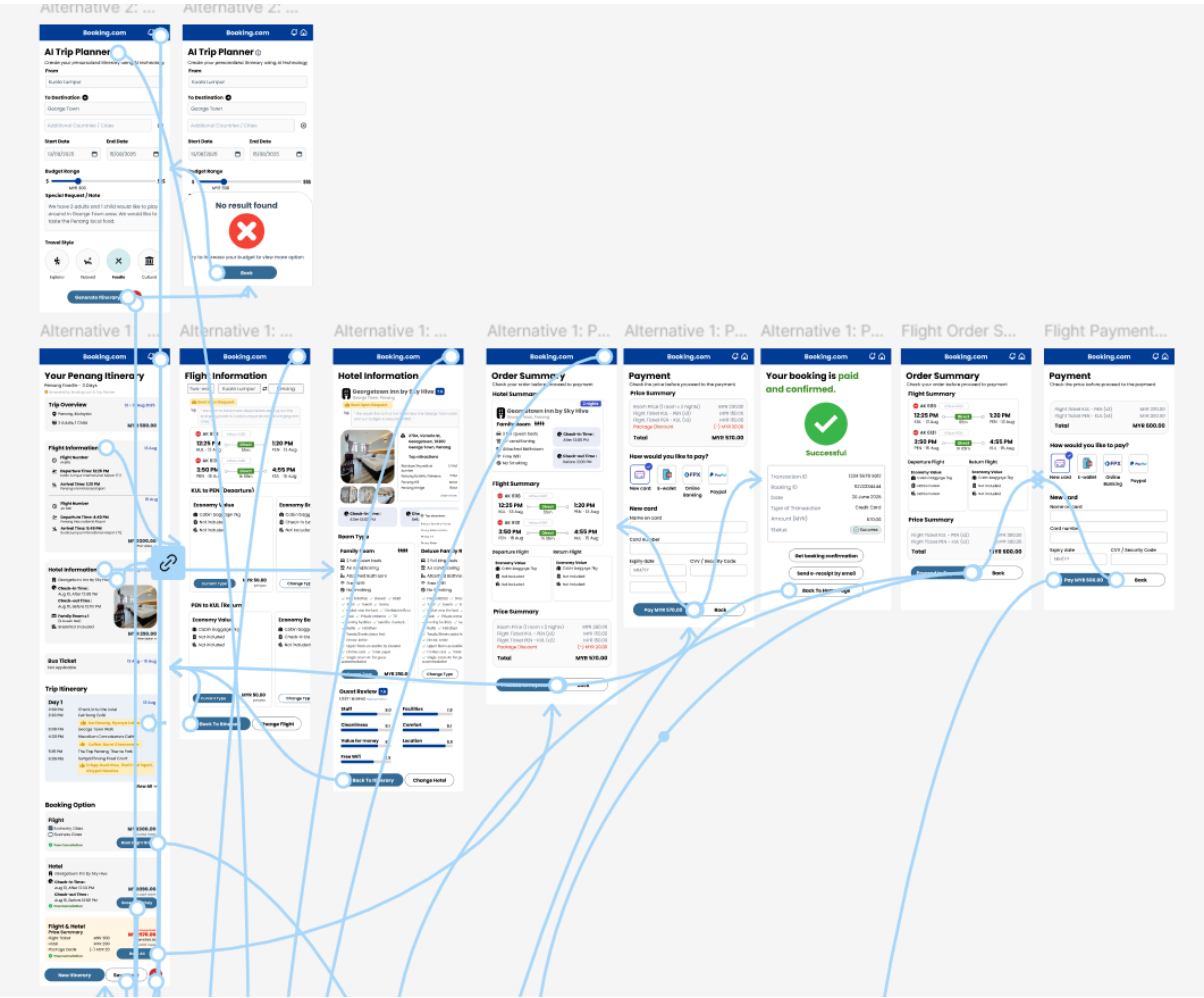
This booking confirmation page will display what the user had booked. They can look back on their booking and get more details and manage their booking.

Draft



This is the draft page where the AI generated itinerary can be saved and other items that they wanted to save.

Alternative for AI Trip Planner

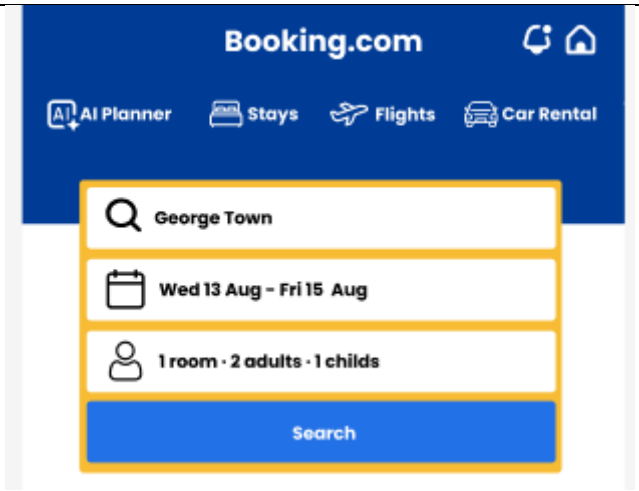


This is the alternative for AI trip planner when the user is provided too low budget for a trip and unsatisfied on the itinerary provided by AI.

3.0 Testing Process

3.1 User Interface Design

3.1.1 Design a transparent interface



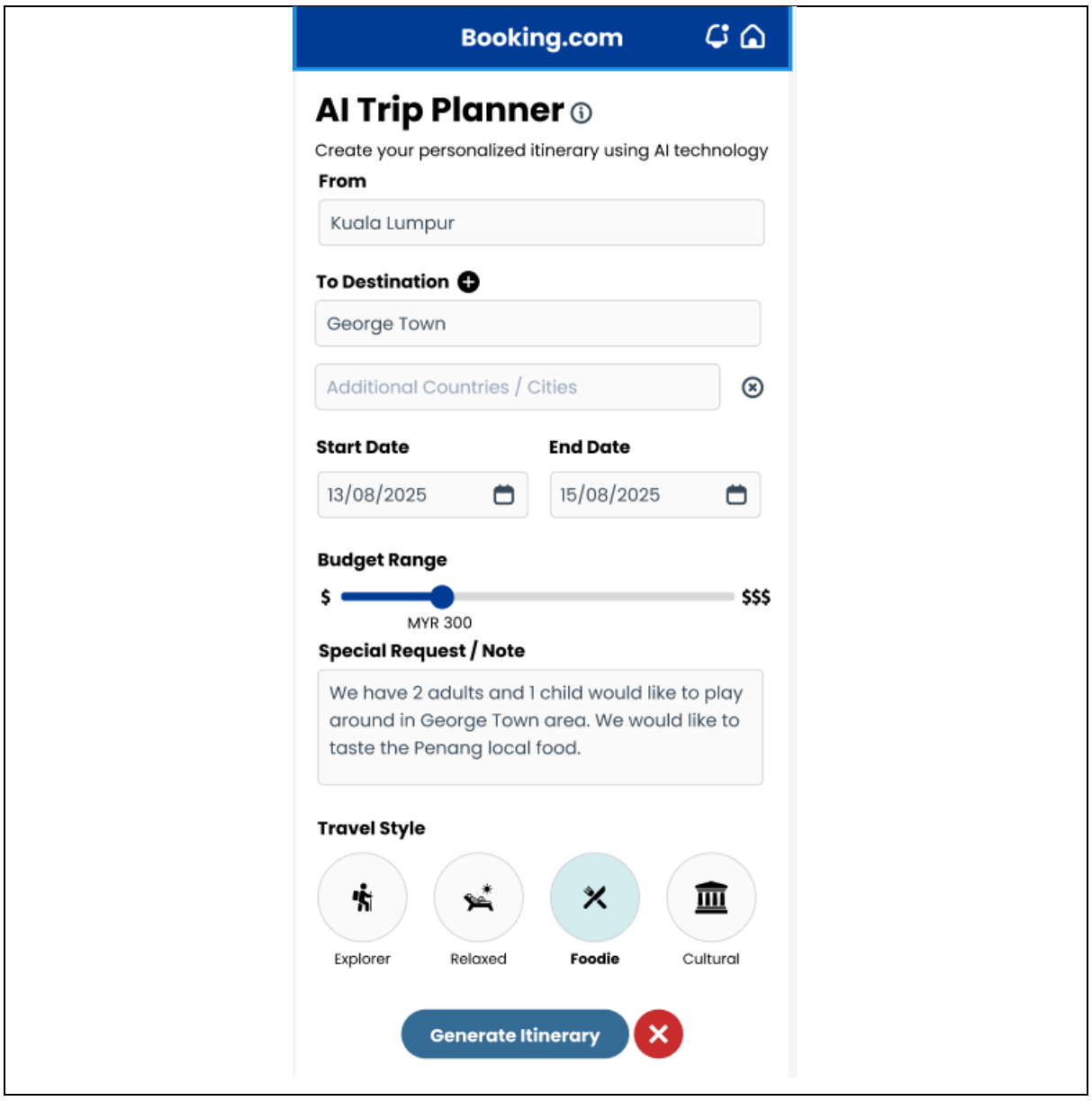
From our interface, we facilitate system design objectives with a clean and focused layout with key functions such as AI Planner, Stays, Flights and more to prioritize users' needs. These functions are also presented using familiar icons and visual cues to make the interface easier to learn and remember. Our UI interface guides users intuitively through the booking process with step-by-step flow. For example, allow users to enter destination, date and persons and press search to find relevant service. This helps users to improve efficiency and productivity by minimizing unnecessary actions and reducing time spent navigating. Additionally, a consistent placement of elements such as the "Search" button directly below the input fields or the use of icons such as notification and home button at the top right to ensure a predictable function for users.

3.1.2 Create easy to learn and use interface



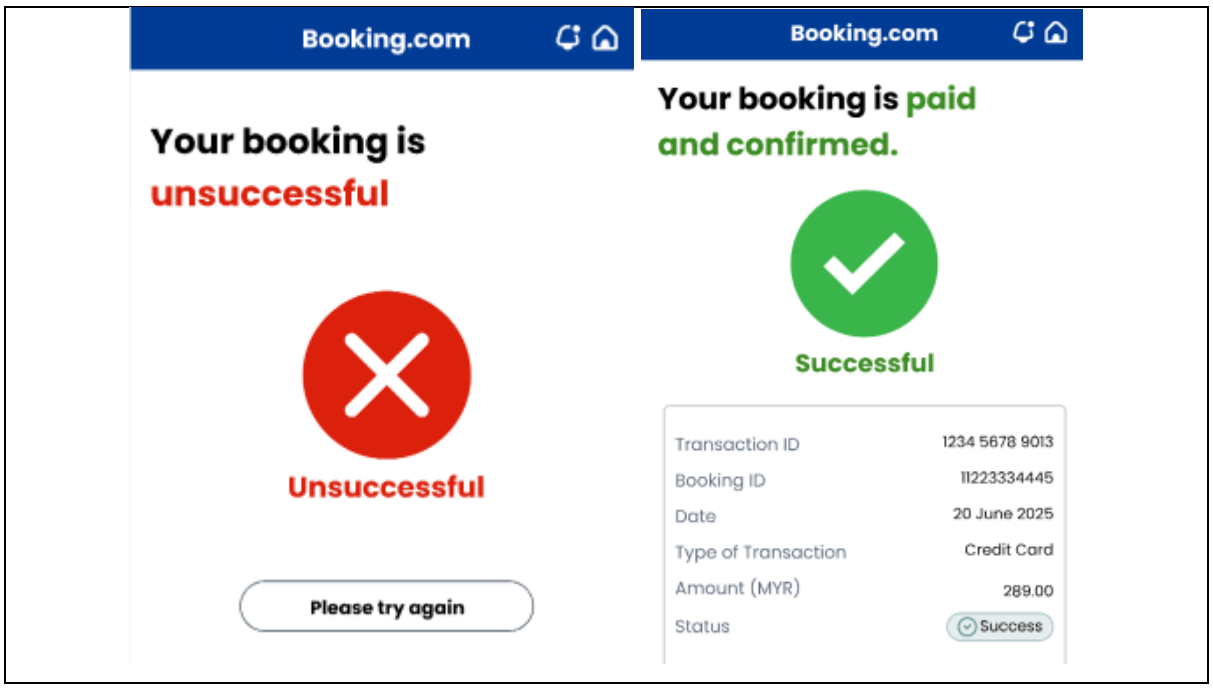
The interface above is our bottom UI, it includes four different functions which are search, draft, my booking and account. The search button navigates users to search their desired planning, draft is where customers have an preliminary version of idea and is unsure of certain part yet, they can save their current version into draft, my booking function which is used to let users check what booking confirmation they have done, and account function which is used for changing profile details, change account and more. These functions are label with both icon and word that users can understand easily, thus making our UI interface easy to navigate.

3.1.3 Minimize input data problems



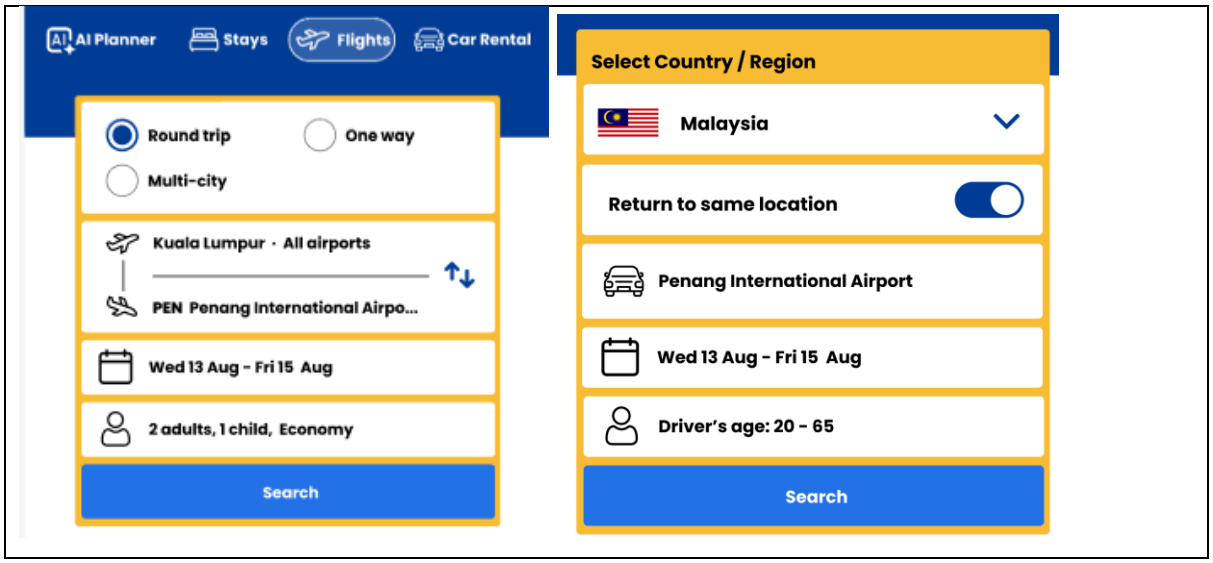
We reduce input data problems by creating fixed inputs for users to fill in to set their plan. An example from the above AI Trip Planner page, it prompts users to enter their destination, date, budge range and travel style and set a plan for users based on this information. Moreover, users need to fill in every requirement to receive a planning, or a notification will show up and remind users to enter that requirement if the system detects that a requirement is lacking.

3.1.4 Provide feedback to users



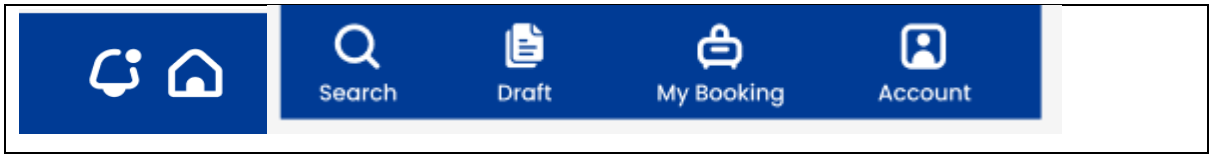
When users are satisfying which certain service and is confirming the payment but the payment fail, this interface on the left will pop up and remain on the screen to notify users that their payment is unsuccessful, users can press on “Please try again” to retry the payment until success. When the payment is successful, the interface on the right will pop up and remain on the screen to notify users that their payment is successful, which are additional details such as their transaction ID, booking ID and more.

3.1.5 Create an attractive layout and design



Our primary color for the layout throughout the whole prototype is blue and white. By limiting the color used for the layout, we reduce users' visual fatigue, which helps maintain a clean and professional appearance, keeping the user's focus on the core content and functionality of the system. Additionally, we highlight the functionality of different services when selected in with white columns as shown from the above interface to indicate the currently selected service. This design choice makes it easier for users to identify which service they are interacting at the moment.

3.1.6 Use familiar terms and images



We use familiar images such as notification icon, home icon, search icon and more as above which is commonly seen in different applications. This helps users to understand the purpose of these functions without the need of additional instructions, thereby making the interface more intuitive for both new and experienced users.

3.2 Pilot Test

The pilot test is done internally before outsourcing for other evaluators. The pilot test ensures that our prototype works smoothly with no errors. Next, we recruit a total of 4 participants to test our prototype, two of the participants are invited though online text, while other two participants are meet in person to test out our prototype.

3.3 Usability Testing

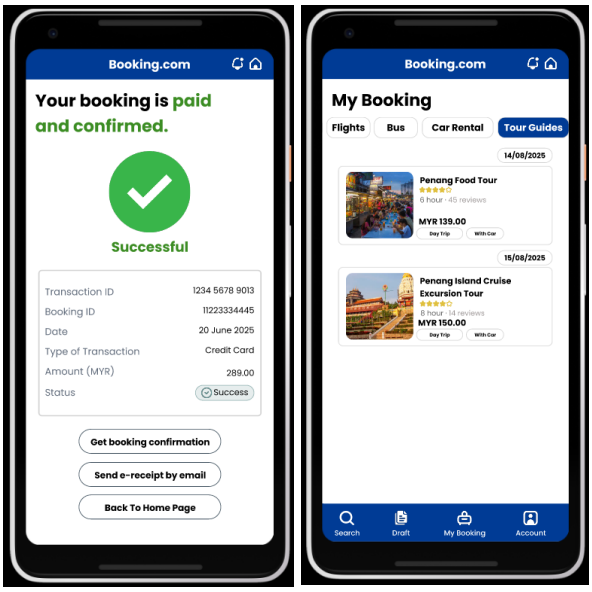
3.3.1 Tour Guide Booking

Tester	Cheong Jun Yuan
Test date	17 July 2025
Prototype developer	Liang Yrenn Da

Test Case 1

Potential Test Inputs (with example)	1. The user selects “Continue with email” as login method. 2. The user enters the email address. 2.1. Example: admin@gmail.com 3. The user enters the verification code 3.1. Example: 123456 4. The user navigates and clicks on the tour guide section 5. The user enters search details: example; 5.1. Destination: George Town 5.2. Duration: 13 August – 15 August 5.3. Attendees: 1 room for 2 adults and 1 child 6. The user clicks the search button 7. The user clicks on Ahmad’s profile 8. The user clicks Book Now 9. The user clicks Proceed to Payment 10. The user enters their card details for payment 11. The user clicks the pay button

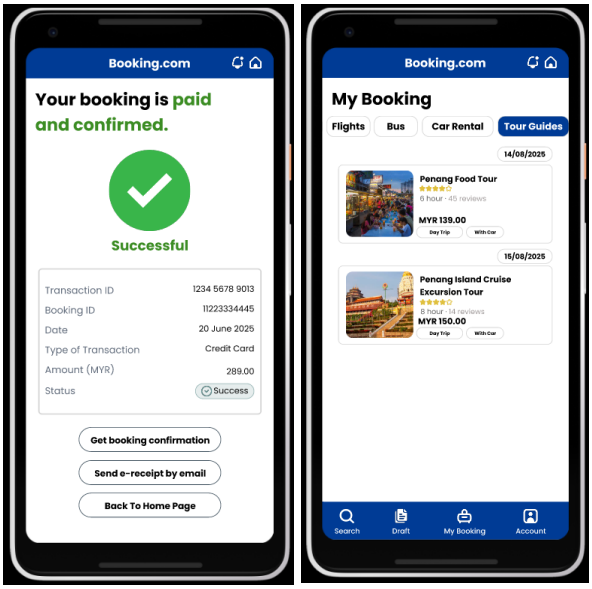
	12. The user clicks the “Get booking confirmation”
Expected Test Outputs (with example)	1. A prompt appears requesting the user enter their email address 2. A prompt appears to enter the verification code sent to the provided email 3. The user is successfully logged in and redirected to the main dashboard/homepage 4. A list or interface for tour guide services is displayed 5. Search form accepts input for destination, duration, and attendees 6. Search results show available tour guides based on search details 7. Ahmad’s tour guide profile opens with details, pricing, availability, and reviews 8. Booking summary page appears with trip details, price, and a “Proceed to Payment” option 9. Payment page opens with fields for card/payment details 10. Card fields are populated correctly; system validates card format 11. Payment is processed. A success message 12. A confirmation page is displayed with booking reference number, details of the tour, and possibly an option to download or email the confirmation
Actual Test Results	1. Users are able to successfully login into their account and access homepage: **Sucessful** 2. User is able to search up list of tour guides: **Sucessful**

	3. User is able to view tour guides infomation: **Sucessful** 4. User is able to confirm booking with tour guide: **Sucessful** 5. User is able to make payment: **Sucessful** 6. User is able to view their booking confirmation**:** **Sucessful** 

Test Case 2

Potential Test Inputs (with example)	1. The user selects “Continue with email” as login method. 2. The user enters the email address. 2.1. Example: admin@gmail.com 3. The user enters the verification code 3.1. Example: 123457 4. The user clicks back on the popup message displaying login failure 5. The user enters the verification code 5.1. Example: 123456 6. The user navigates and clicks on the tour guide section 7. The user enters search details: example;

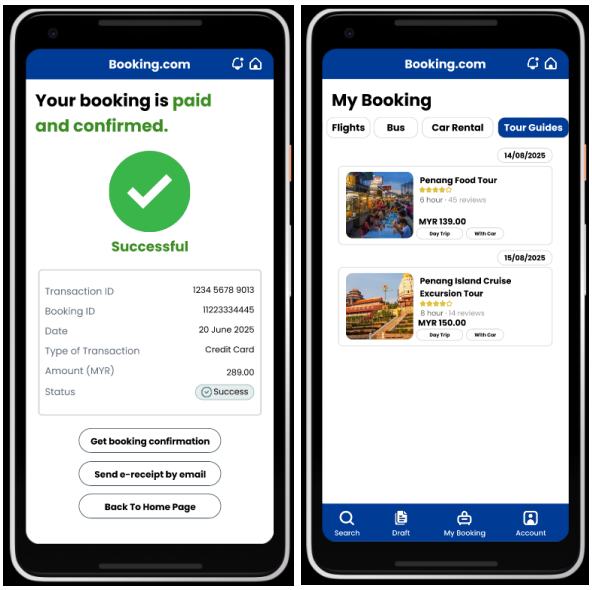
	7.1. Destination: George Town 7.2. Duration: 13 August – 15 August 7.3. Attendees: 1 room for 2 adults and 1 child 8. The user clicks search button 9. The user clicks on Ahmad's profile 10. The user clicks Book Now 11. The user clicks Proceed to Payment 12. The user enters their card details for payment 13. The user clicks the pay button 14. The user clicks the "Get booking confirmation"
Expected Test Outputs (with example)	1. A prompt appears requesting the user enter their email address 2. A prompt appears to enter the verification code sent to the provided email 3. Login failure popup is displayed due to incorrect verification code 4. User is returned to the verification code entry screen 5. The user is successfully logged in and redirected to the main dashboard/homepage 6. A list or interface for tour guide services is displayed 7. Search form accepts input for destination, duration, and attendees 8. Search results show available tour guides based on search details 9. Ahmad's tour guide profile opens with details, pricing, availability, and reviews 10. Booking summary page appears with trip details, price, and a "Proceed to Payment" option

	11. Payment page opens with fields for card/payment details 12. Card fields are populated correctly; system validates card format 13. Payment is processed. A success message 14. A confirmation page is displayed with booking reference number, details of the tour, and possibly an option to download or email the confirmation
Actual Test Results	1. Users are able to successfully login into their account after login failure: **Successful** 2. User is able to search up list of tour guides: **Successful** 3. User is able to view tour guides: **Successful** 4. User is able to confirm booking with tour guide: **Successful** 5. User is able to make payment: **Successful** 6. User is able to view their booking confirmation**:** **Successful** 

Test Case 3

Potential Test Inputs (with example)	1. The user selects “Continue with email” as login method. 2. The user enters the email address. 2.1. Example: admin@gmail.com 3. The user enters the verification code 3.1. Example: 123456 4. The user navigates and clicks on the tour guide section 5. The user enters search details: example; 5.1. Destination: George Town 5.2. Duration: 13 August – 15 August 5.3. Attendees: 1 room for 2 adults and 1 child 6. The user clicks search button 7. The user clicks on Ahmad’s profile 8. The user clicks back to listing 9. The user selects another one 9.1. Example: Nurul 10. The user clicks Book Now 11. The user clicks Proceed to Payment 12. The user enters their card details for payment 13. The user clicks the pay button 14. The user clicks the “Get booking confirmation”
Expected Test Outputs (with example)	1. A prompt appears requesting the user enter their email address 2. A prompt appears to enter the verification code sent to the provided email 3. The user is successfully logged in and redirected to the main dashboard/homepage 4. A list or interface for tour guide services is displayed

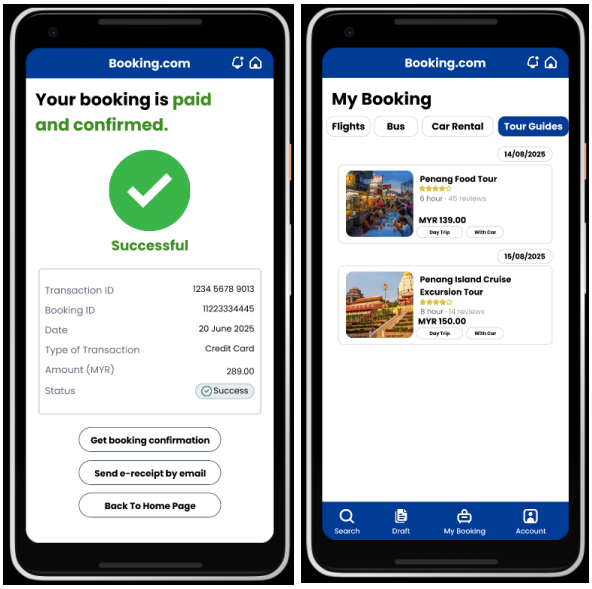
	5. Search form accepts input for destination, duration, and attendees 6. Search results show available tour guides based on search details 7. Ahmad's tour guide profile opens with details, pricing, availability, and reviews 8. User is returned to the list of available tour guides 9. Nurul tour guide profile opens with details, pricing, availability, and reviews 10. Booking summary page appears with trip details, price, and a "Proceed to Payment" option 11. Payment page opens with fields for card/payment details 12. Card fields are populated correctly; system validates card format 13. Payment is processed. A success message 14. A confirmation page is displayed with booking reference number, details of the tour, and possibly an option to download or email the confirmation
Actual Test Results	1. Users are able to successfully login into their account: **Sucessful** 2. User is able to search up list of tour guides: **Sucessful** 3. User is able to view tour guides: **Sucessful** 4. User is able to view other tour guides: **Sucessful** 5. User is able to confirm booking with tour guide: **Sucessful** 6. User is able to make payment: **Sucessful**

	7. User is able to view their booking confirmation: **Sucessful** 

Test Case 4

Potential Test Inputs (with example)	1. The user selects “Continue with email” as login method. 2. The user enters the email address. 2.1. Example: admin@gmail.com 3. The user enters the verification code 3.1. Example: 123456 4. The user navigates and clicks on the tour guide section 5. The user enters search details: example; 5.1. Destination: George Town 5.2. Duration: 13 August – 15 August 5.3. Attendees: 1 room for 2 adults and 1 child 6. The user clicks search button 7. The user clicks on Ahmad’s profile 8. The user clicks Book Now 9. The user clicks Proceed to Payment 10. The user enters their card details for payment

	11. The user clicks the pay button 12. The user clicks back to go back to card details 13. The user enters their correct card details for payment 14. The user clicks the pay button 15. The user clicks the “Get booking confirmation”
Expected Test Outputs (with example)	1. A prompt appears requesting the user enter their email address 2. A prompt appears to enter the verification code sent to the provided email 3. The user is successfully logged in and redirected to the main dashboard/homepage 4. A list or interface for tour guide services is displayed 5. Search form accepts input for destination, duration, and attendees 6. Search results show available tour guides based on search details 7. Ahmad’s tour guide profile opens with details, pricing, availability, and reviews 8. Booking summary page appears with trip details, price, and a “Proceed to Payment” option 9. Payment page opens with fields for card/payment details 10. Card fields are entered; system validates card format 11. Payment is processed. An error message appears. 12. User is returned to the card details entry screen

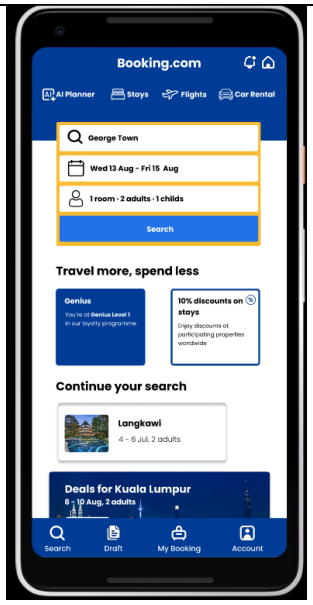
	13. Card fields are entered correctly; system validates card format 14. A confirmation page is displayed with booking reference number, details of the tour, and possibly an option to download or email the confirmation
Actual Test Results	1. Users are able to successfully login into their account: **Successful** 2. User is able to search up list of tour guides: **Successful** 3. User is able to view tour guides: **Successful** 4. User is able to confirm booking with tour guide: **Successful** 5. User is able to make payment after error: **Successful** 6. User is able to view their booking confirmation**:** **Successful** 

3.3.2 Manual Booking

Tester	Vincci Chin Wen Xi
Test date	17 July 2025
Prototype developer	Liang Yrenn Da

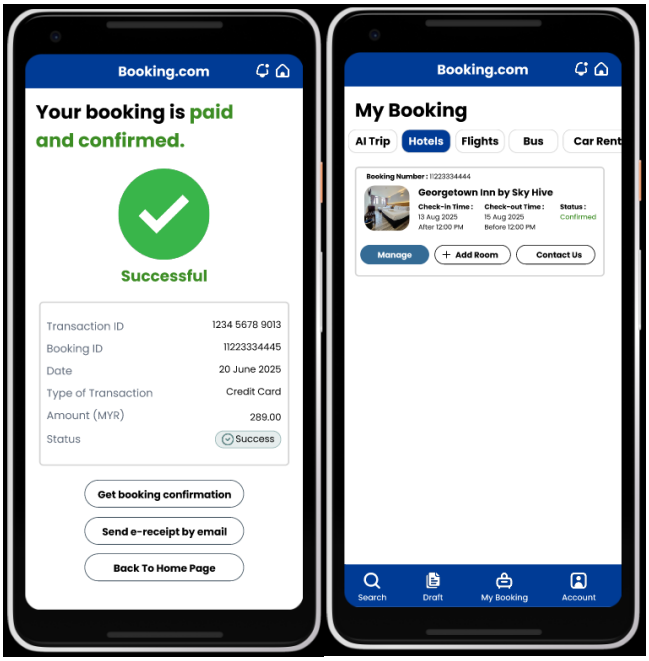
Test Case 1

Potential Test Inputs (with example)	1. The user opens Booking.com Application. 2. The user selects “Continue with email” as login method. 3. The user enters the email address. 4. The user receives the verification code from email and fills in the code. 5. The user verifies the email to login. (EX1: Login failure, try to get the new code and verify again until successful) 6. The user is on the “home page”.
Expected Test Outputs (with example)	1. The user opens Booking.com Application. 2. The user selects “Continue with email” and fill their email. 3. The user receives the verification and fills in the code. 4. The user verifies the email to login. 5. The user is on the “home page”.
Actual Test Results	1. The user opens Booking.com Application: **Successful** 2. The user selects “Continue with email” and fill their email: **Successful** 3. The user receives the verification and fills in the code: **Successful** 4. The user verifies the email to login: **Successful** 5. The user is on the “home page”: **Successful**

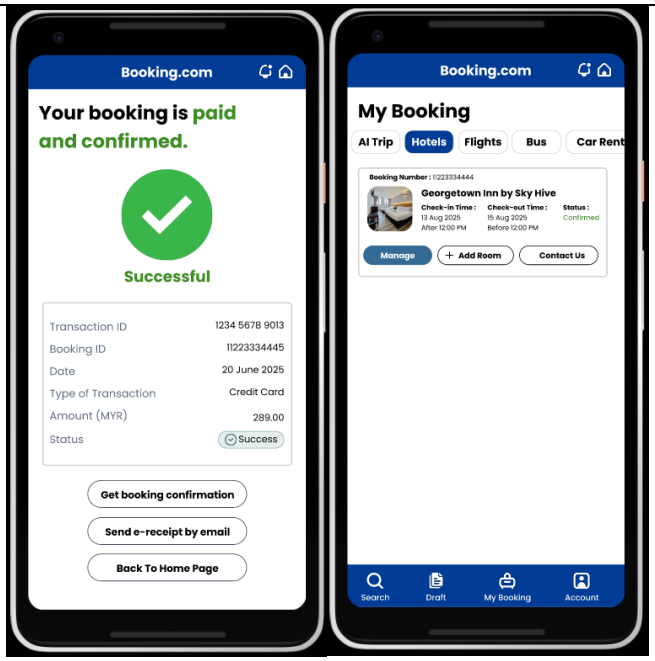
Test Case 2

Potential Test Inputs (with example)	1. The user opens the home page 2. The user clicks on “stays: to book hotel 3. The user searches for hotel by filling details: Location: George Town Date: 13 Aug – 15 Aug Type: 1 room – 2 adults – 1 child 4. The user clicks search to find suitable hotel 5. The user selects the hotel available 6. The user will proceed to review the hotel summary 7. The user will proceed to payment window 8. The user will choose their payment method 9. The payment is successful 10. The user gets booking confirmation
Expected Test Outputs (with example)	1.The user opens the home page 2.The user clicks on “stays: to book hotel 3.The user searches for the hotel by filling in details: Location: George Town

	Date: 13 Aug – 15 Aug Type: 1 room – 2 adults – 1 child 4.The user clicks search to find suitable hotel 5.The user selects the hotel available 6.The user will proceed to review the hotel summary 7.The user will proceed to payment window 8.The user will choose their payment method 9.The payment is successful 10.The user gets to booking confirmation
Actual Test Results	1. The user opens the homepage: **Successful** 2. The user selects “stays” to book hotel: **Successful** 3. The user clicks search to review hotel available**: Successful** 4. The user selects book now: **Successful** 5. The user proceed payment: **Successful** 6. The user receives booking information: **Successful** 

Test Case 3

Potential Test Inputs (with example)	1. The user opens the home page 2. The user selects “stay” to proceed to hotel booking 3. The user selects the hotel from past searches 4. The user will proceed to review the hotel summary 5. The user will proceed to payment window 6. The user will choose their payment method 7. The payment is successful 8. The user gets to booking confirmation
Expected Test Outputs (with example)	1. The user opens the home page 2. The user selects “stay” to proceed to hotel booking 3. The user selects the hotel from past searches 4. The user will proceed to review the hotel summary 5. The user will proceed to payment window 6. The user will choose their payment method 7. The payment is successful 8. The user gets booking confirmation
Actual Test Results	1. The user opens the homepage: **Successful** 2. The user selects “stays” to book hotel: **Successful** 3. The user clicks on past hotel searches to book**:** **Successful** 4. The user selects book now: **Successful** 5. The user proceed payment: **Successful** 6. The user receives booking information: **Successful**

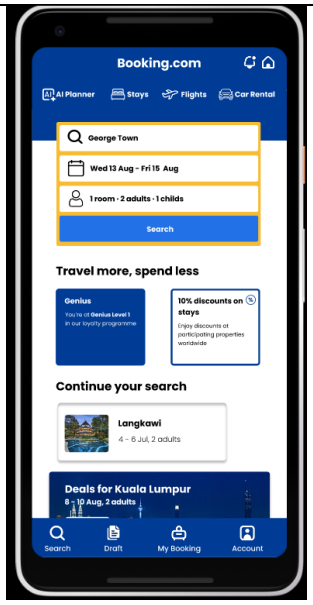

Booking.com
Your booking is paid and confirmed.
Successful
Transaction ID 1234 5678 9013
Booking ID 11223334445
Date 20 June 2025
Type of Transaction Credit Card
Amount (MYR) 289.00
Status Success
Get booking confirmation
Send e-receipt by email
Back To Home Page
Booking.com
My Booking
AI Trip
Hotels
Flights
Bus
Car Rent
Georgetown Inn by Sky Hive
Manage
Add Room
Contact Us
Search
Draft
My Booking
Account

3.3.3 Transportation Booking

Tester	Isabelle Yau Li Shuang
Test date	17 July 2025
Prototype developer	Ooi Jin Yon

Test Case 1

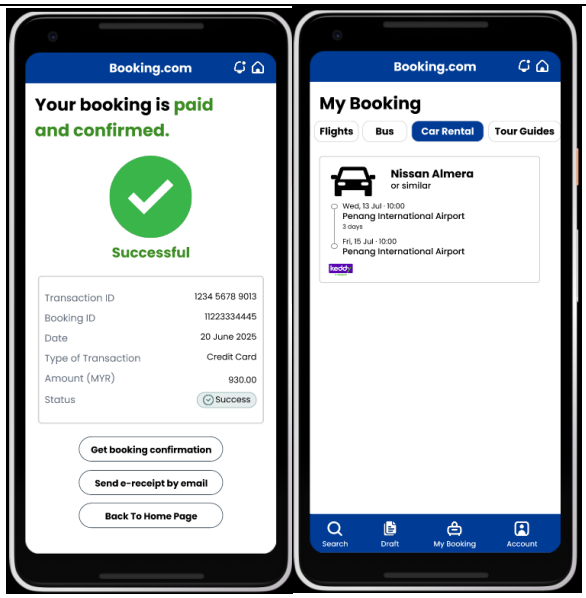
Potential Test Inputs (with example)	1. The user opens Booking.com Application. 2. The user selects “Continue with email” as login method. 3. The user enters the email address. 4. The user receives the verification code from email and fills in the code. 5. The user verifies the email to login. (**EX1**: Login failure, try to get the new code and verify again until successful) 6. The user is on the “home page”.
Expected Test Outputs (with example)	1. The user opens Booking.com Application. 2. The user selects “Continue with email” and fill their email. 3. The user receives the verification and fills in the code. 4. The user verifies the email to login. 5. The user is on the “home page”.
Actual Test Results	1. The user opens Booking.com Application: **Successful** 2. The user selects “Continue with email” and fill their email: **Successful** 3. The user receives the verification and fills in the code: **Successful** 4. The user verifies the email to login: **Successful** 5. The user is on the “home page”: **Successful**

Test Case 2

Potential Test Inputs (with example)	1.The user is on “Car Rental” 2.They search for buses by filling in the details: Pick up location: “Penang International Airport” Duration: “13 Aug – 15 Aug” Driver’s age: “20-65”. 3.They click on “Search” button. 4.They browse and select a suitable rental car. 5.They proceed to the specific rental car information page. 6.After they view the full information, they will proceed to payment. 7.Before payment, they can choose to add protection or not. 8.They will look through the booking summary 9.They will proceed with the payment process 10.They choose their payment method 11.Their payment is successful. (EX1: The payment may be unsuccessful, and need make the payment again until successful) 12. Back to booking confirmation page.

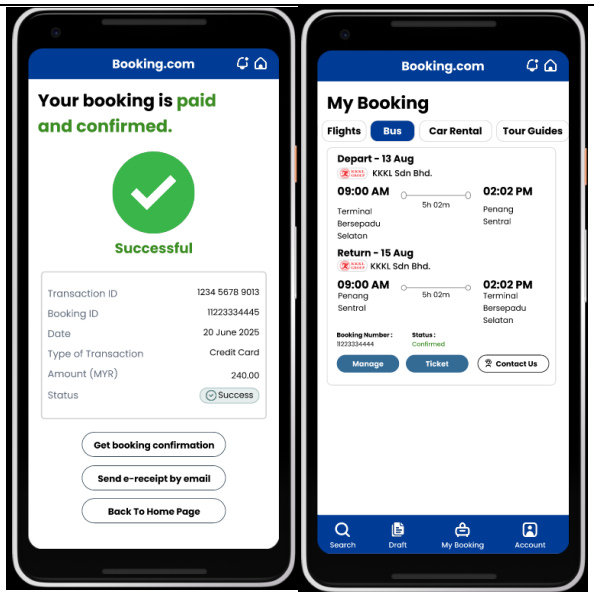
Expected Test Outputs (with example)	1. The user is on “Car Rental” 2. They are searching for a rental car that fits their requirements. 3. The available car rental option has been listing out for them to choose the most suitable. Duration: 13 Aug – 15 Aug Pick-up location: Penang International Airport 4. A few suitable car rental options will be shown, and the user will choose the most suitable from below: Duration: 13 Aug – 15 Aug Type of Car: Nissan Almera, Proton Persona, Honda HR-V, Nissan X-Trail. 5. The user selects the car and proceeds to payment 6. The user selects to add on the travel protection or not. 7. The user selects a suitable payment method: New Card, E-wallet, Online banking, PayPal 8. They successfully book their rental car on 13 Aug – 15 Aug in Penang International Airport.
Actual Test Results	1. The user in on “Car Rental”: **Successful** 2. They selected a suitable rental car based on the requirements: **Successful** 3. They selected the addon travel protection insurance: **Successful** 4. They select their payment method and succeed in making their payment: **Successful** 5. They go back to homepage or redirect to booking confirmation page: **Successful**

Test Case 3

Potential Test Inputs (with example)	1. The user is on “Bus Ticket” 2. They chose their bus option, either round trip or one way. 3. They search for buses by filling in the details. Departure: “Terminal Bersepadu Selatan” Arrival: “Penang Sentral (Butterworth)” Number of persons: “2 adults, 1 child” 4. They browse and select a suitable depart bus ticket. 5. They will choose their seat 6. They browse and select a suitable return bus ticket. 7. They will choose their seat 8. They will view their order summary 9. After they view the summary, they will go to the payment page. 10. They choose their payment method 11. Their payment is successful. (EX1: The payment may be unsuccessful, and need make the payment again until successful) 12. Back to booking confirmation page.

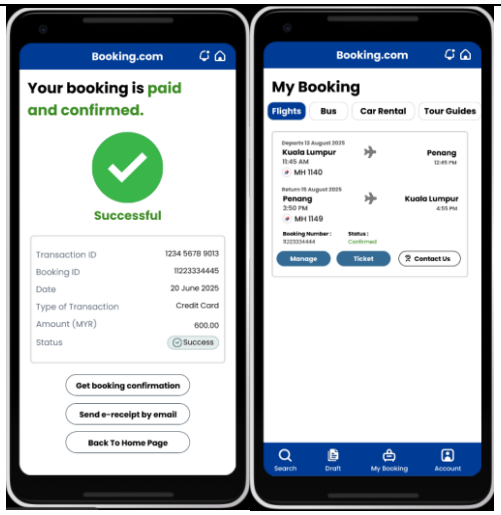
Expected Test Outputs (with example)	1. The user is on “Bus ticket” 2. They are searching for a bus ticket that fits their requirements. 3. The available bus ticket has been listing out for them to choose the most suitable. Departure: “Terminal Bersepadu Selatan” Arrival: “Penang Sentral (Butterworth)” Number of persons: “2 adults, 1 child” 4. They click on “Search” button. 5. The user selects the bus and picks their seat for a two-way ticket. 6. They proceed to payment page 7. The user selects a suitable payment method: New Card, E-wallet, Online banking, PayPal 8. They successfully booked their two-way bus ticket from TBS to Penang Sentral.
Actual Test Results	1. The user is on “Bus ticket”: **Successful** 2. They selected a suitable bus ticket based on the requirements: **Successful** 3. They selected their seat: **Successful** 4. They selected their payment method and succeeded in making their payment: **Successful** 5. They go back to homepage or redirect to booking confirmation page: **Successful**

Test Case 4

Potential Test Inputs (with example)	1. The user is on “Flight” 2. They chose their flight option, either round trip or one way. 3. They search for flight by filling in the details. **Departure**: “Kuala Lumpur – All Airports” **Arrival**: “Penang International Airport” **Number of persons**: “2 adults, 1 child” 4. They browse and select a suitable flight. 5. They will choose their desired time. 6. They browse and select a suitable return flight. 7. They will choose their desired time 8. They will view their order summary 9. After they view the summary, they will go to the payment page. 10. They choose their payment method 6. Their payment is successful. (**EX1**: The payment may be unsuccessful, and need make the payment again until successful) 7. Back to booking confirmation page.

Expected Test Outputs (with example)	1. The user is on “Flight” 2. They are searching for a flight that fits their requirements. 3. The available flight has been listing out for them to choose the most suitable. **Departure**: “Kuala Lumpur – All Airports” **Arrival**: “Penang International Airport” **Number of persons**: “2 adults, 1 child” 4. They click on “Search” button. 5. The user selects the flight with the desired time for a two-way ticket. 6. They proceed to payment page 7. The user selects a suitable payment method: New Card, E-wallet, Online banking, PayPal 8. They successfully booked their two-way flight ticket from Kuala Lumpur – All Airports to Penang International Airport.
Actual Test Results	1. The user is on “Flight”: **Successful** 2. They selected a suitable flight based on the requirements: **Successful** 3. They selected their desired time: **Successful** 4. They selected their payment method and succeeded in making their payment: **Successful** 5. They go back to homepage or redirect to booking confirmation page: **Successful**

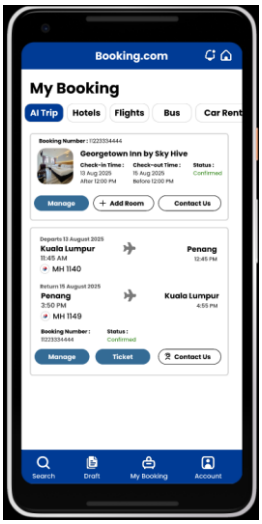
	

3.3.4 AI Trip Planner

Tester	Peggy Liow En Ki
Test date	14 July 2025
Prototype developer	Ooi Jin Yon, Liang Yrenn Da, Yeoh Jie Long

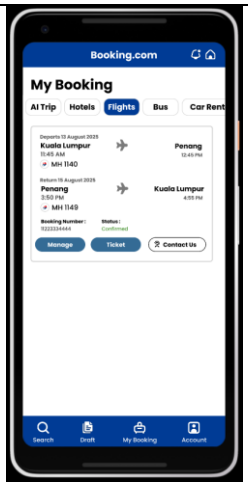
Test Case 1

Potential Test Inputs (with example)	1. User accesses the “AI Trip Planner” module. 2. Trip details are input: • From: Kuala Lumpur • To: George Town • Date: 13/8/2025 to 15/8/2025 • Budget: RM1000 • Group: 2 Adults, 1 Child • Travel Style: Foodie • Special Notes: "Play around in George Town and try Penang food" 3. User clicks “Start Generate Itinerary”.
Expected Test Outputs (with example)	1. Itinerary is successfully generated. 2. It includes: • Flight info: e.g., Flight AK6123, RM320 total

	• Hotel info: e.g., Evergreen Laurel Hotel, RM420 total • Trip itinerary for Day 1 and Day 2 with food stops and activities • Booking options shown: [Book Flight Only], [Book Food Only], [Book All] 3. User selects “Book All”. 4. Display of: • Flight summary • Hotel summary • Price summary (RM980) 5. User proceeds to payment and completes transaction. 6. Redirected to Booking Confirmation Page.
Actual Test Results	1. Inputs accepted and itinerary generated: **Successful** 2. Flight, hotel, and itinerary details accurate: **Successful** 3. “Book All” function worked and showed full summary: **Successful** 4. Payment completed and confirmation displayed: **Successful** 

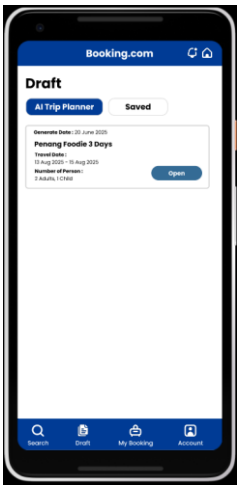
Test Case 2

Potential Test Inputs (with example)	1. User accesses the “AI Trip Planner” module. 2. Trip details are input: • From: Kuala Lumpur • To: George Town • Date: 13/8/2025 to 15/8/2025 • Budget: RM1000 • Group: 2 Adults, 1 Child • Travel Style: Foodie • Special Notes: "Play around in George Town and try Penang food" 3. User clicks “Start Generate Itinerary”.
Expected Test Outputs (with example)	1. Itinerary is successfully generated. 2. It includes: • Flight info: e.g., Firefly FY1004, RM300 total • Hotel info (shown but not booked) 3. Trip itinerary with food spots Booking options: [Book Flight Only], [Book Food Only], [Book All] 4. User selects “Book Flight Only”. 5. Display of: • Flight summary only • Total price: RM300 6. User proceeds to payment. 7. Confirmation page displays only flight booking.
Actual Test Results	1. Itinerary generated: **Successful** 2. Flight details accurate: **Successful** 3. “Book Flight Only” option showed correct summary: **Successful** 4. Payment processed and booking confirmed: **Successful**

Test Case 3

Potential Test Inputs (with example)	1) User accesses the “AI Trip Planner” module. 2) Trip details are input: • From: Kuala Lumpur • To: George Town • Date: 13/8/2025 to 15/8/2025 • Budget: RM1000 • Group: 2 Adults, 1 Child • Travel Style: Foodie • Special Notes: "Play around in George Town and try Penang food" 3) User clicks “Start Generate Itinerary”.
Expected Test Outputs (with example)	1. Itinerary is successfully generated. 2. It includes: • Flight info: e.g., Firefly FY1004, RM300 total • Hotel info (shown but not booked) 3. Trip itinerary with food spots Booking options: [Book Flight Only], [Book Food Only], [Book All] 4. User selects “Save Draft”. 5. Display of:

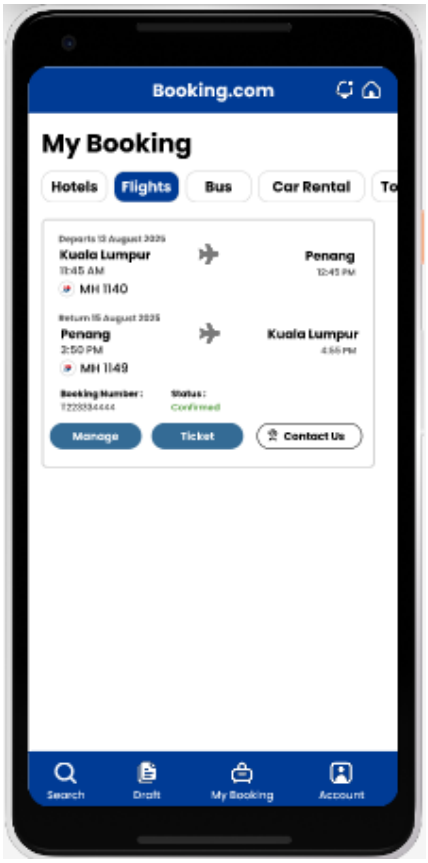
	• Penang Foodie 3 Days • Travel Date • Number of Person
Actual Test Results	1. Itinerary generated: **Successful** 2. Flight details accurate: **Successful** 3. “Save Draft” option showed correct summary: **Successful** 4. Display of AI draft saved: **Successful** 

4.0 Final Prototype

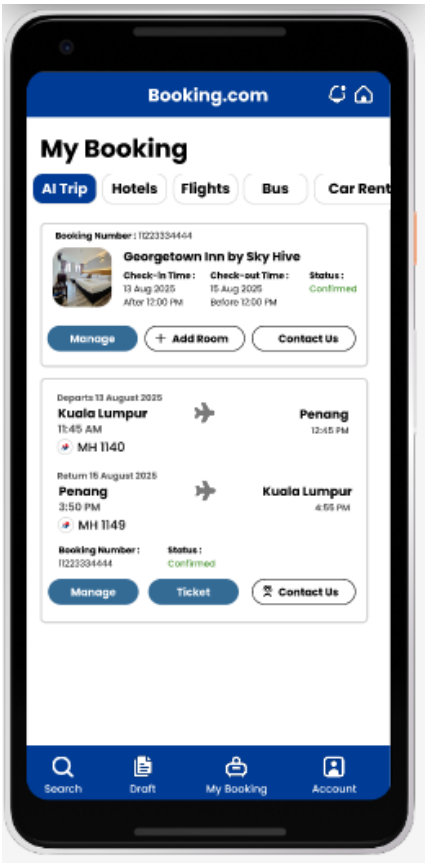
https://www.figma.com/proto/T9G2M3j9EBfy3iVvxAWW8L/BIS2102-Booking.com?node-id=0-1&t=u94DnACMq8xteTRg-1

After receiving feedback from our test users, we make some minor changes in certain page.

Before

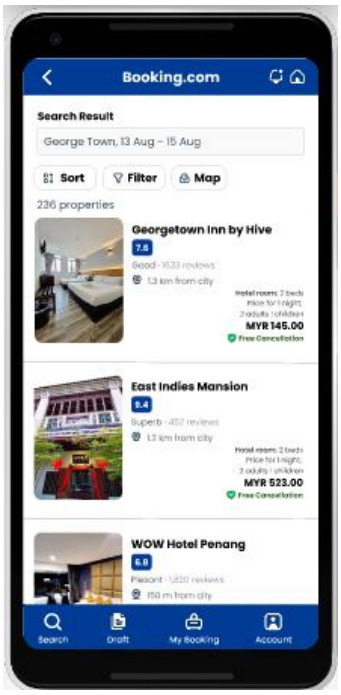


After

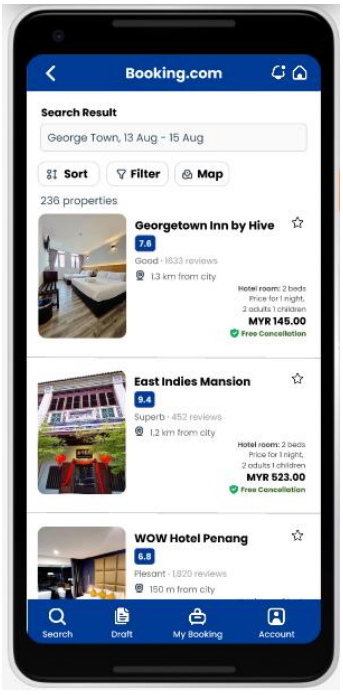


In our booking confirmation page, we follow the feedback to include ‘AI Trip’ section for users when a payment had been done using AI Trip Planner.

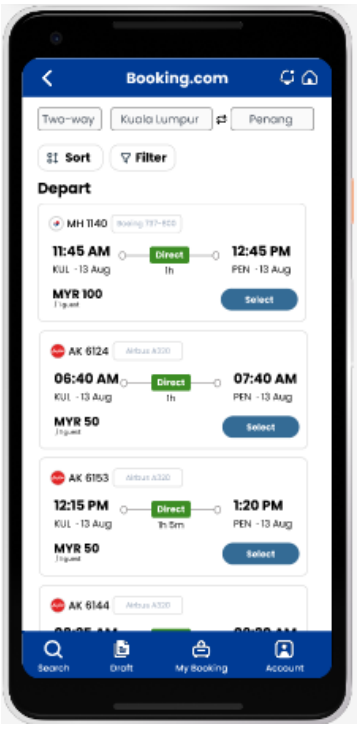
Before



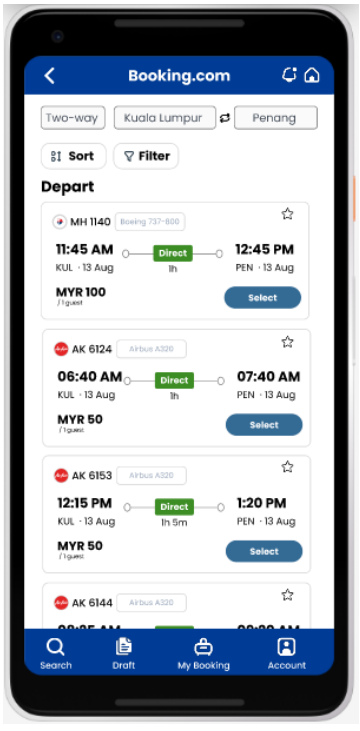
After

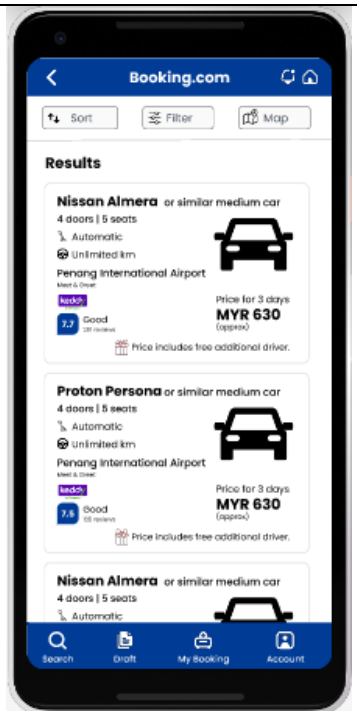


Before

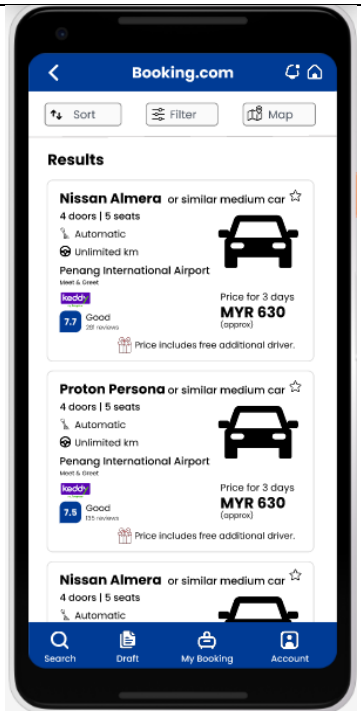


After

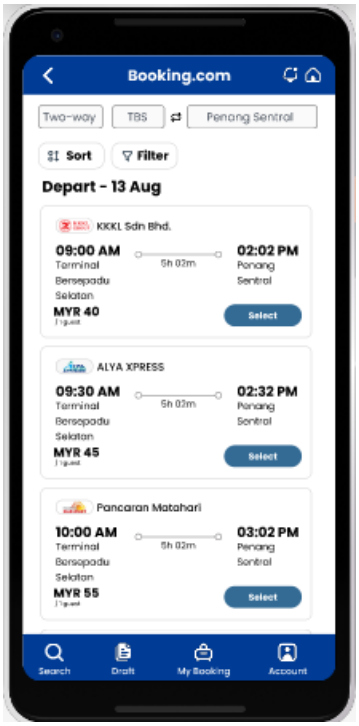




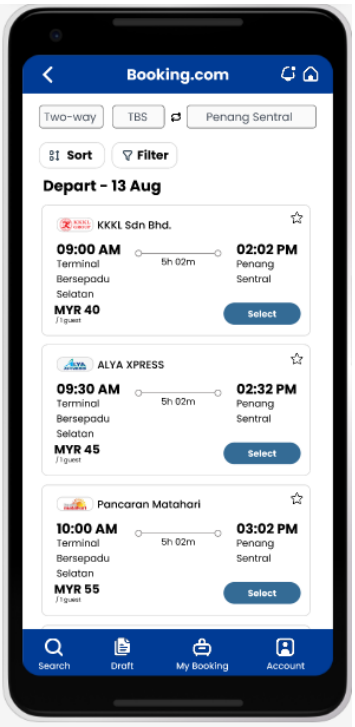
Before



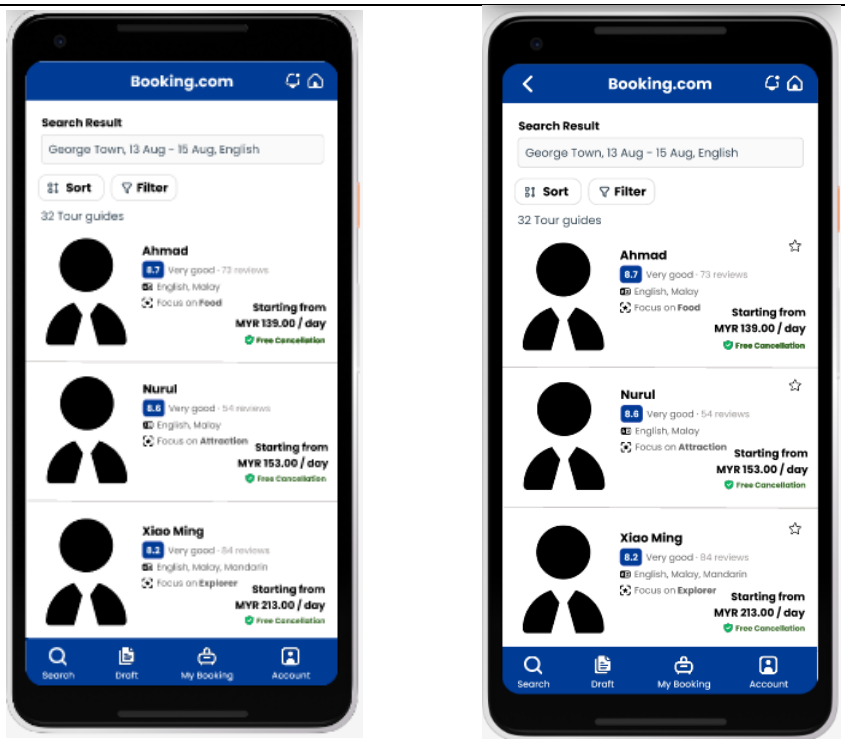
After



Before



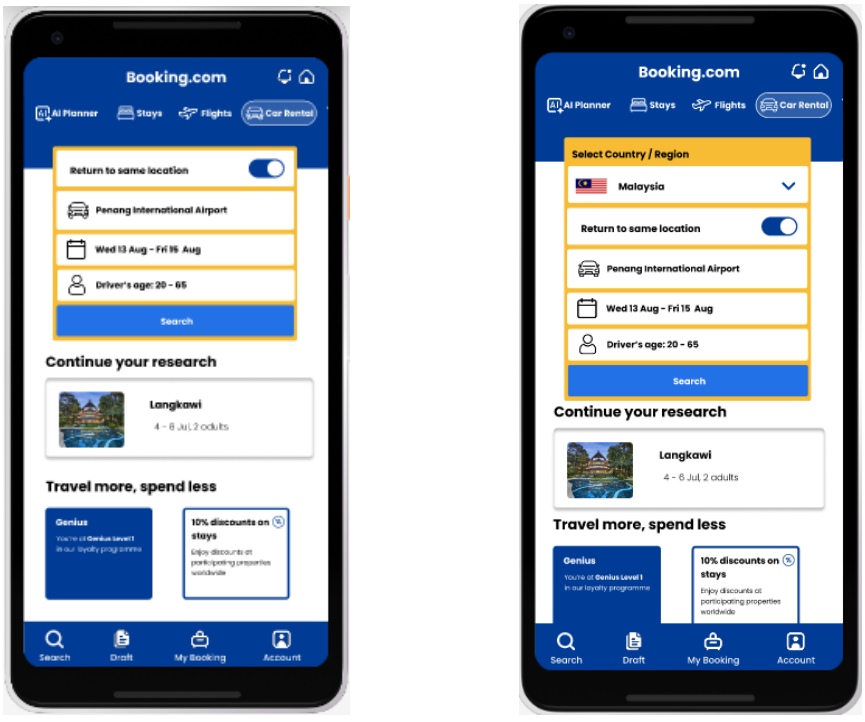
After



We follow the advice from the feedback of adding star icon as a function to save the particular service for later used.

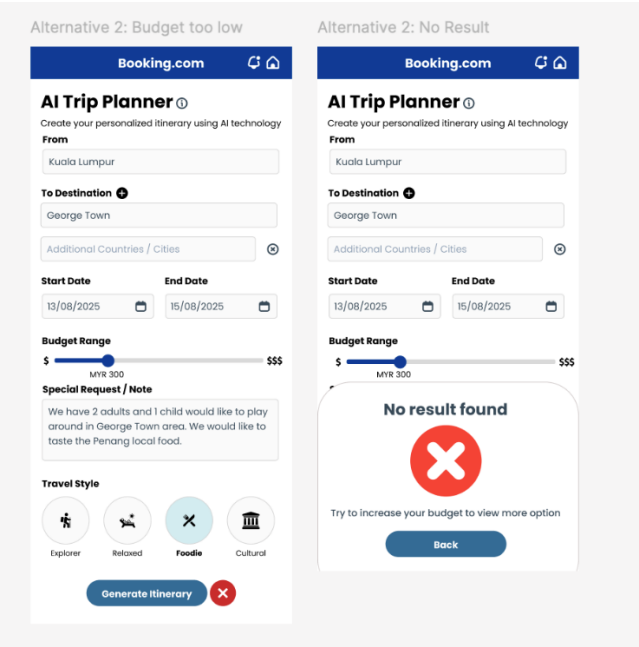
Before

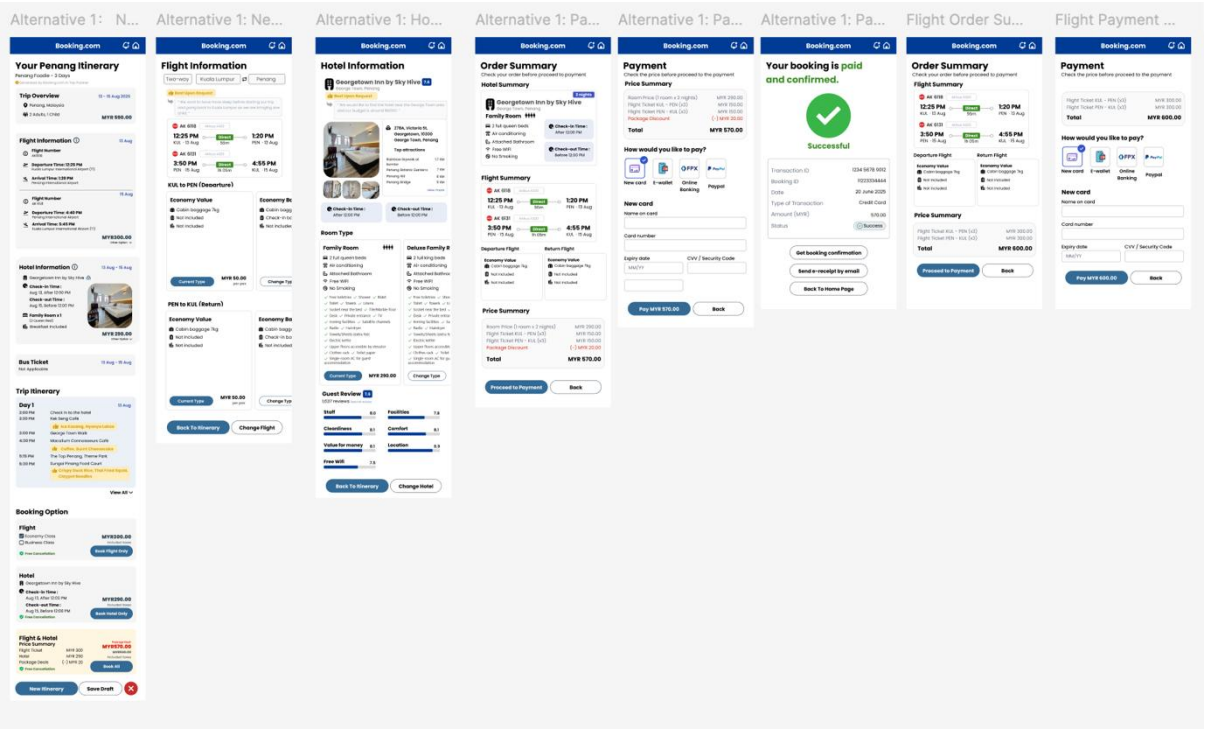
After



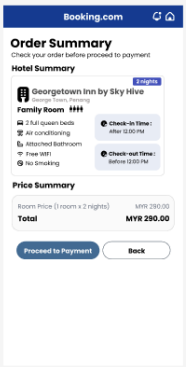
We improve the search functionality of car rental by adding region options for users to filter out unnecessary region and only provide car rental for specific area.

5.0 Digital Mockup

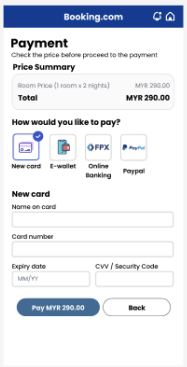




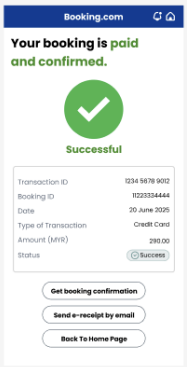
Hotel Order Su...



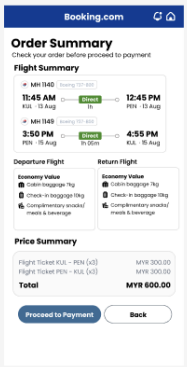
Hotel Payment ...



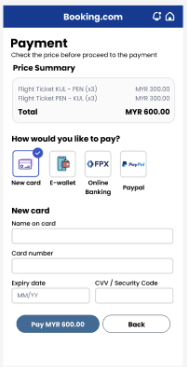
Hotel Purchase ...



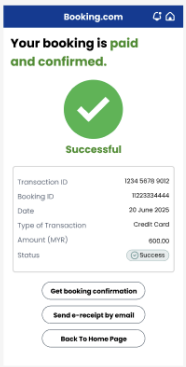
Flight Order Su...



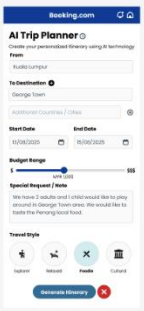
Flight Payment ...



Flight Purchase...



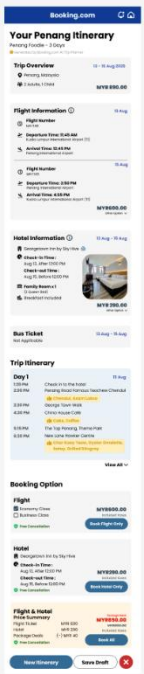
AI Trip Planner ...



AI Trip Planner I...



AI Trip Planner ...



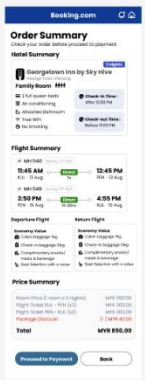
Flight Information



Hotel Information



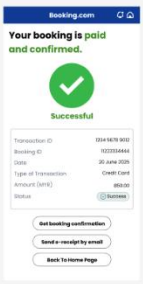
Package Order ...



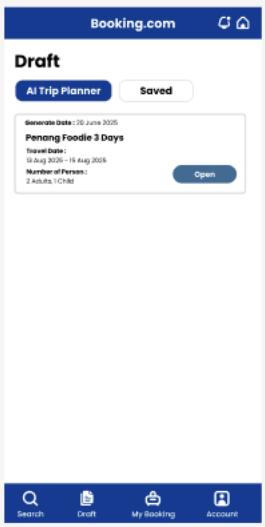
Package Paym...



Package Purch...



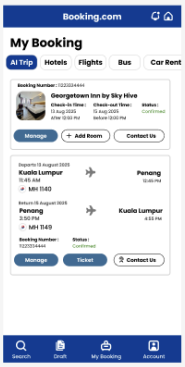
Draft - AI Planner



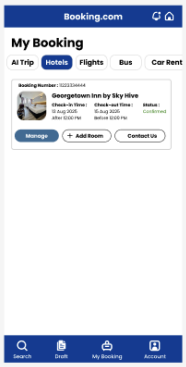
Draft - Saved



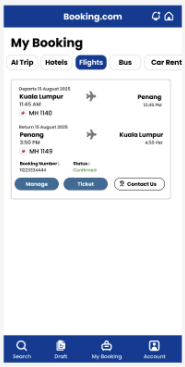
Booking Confir...



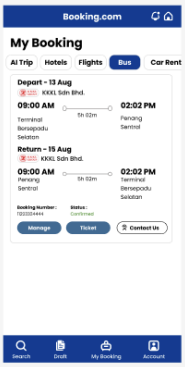
Booking Confir...



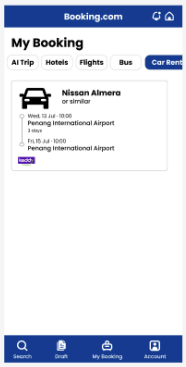
Booking Confir...



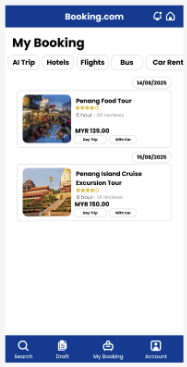
Booking Confir...



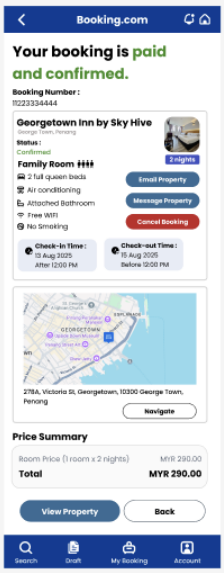
Booking Confir...



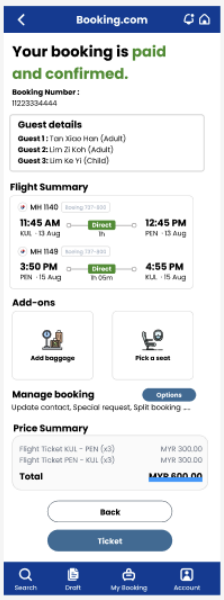
Booking Confir...



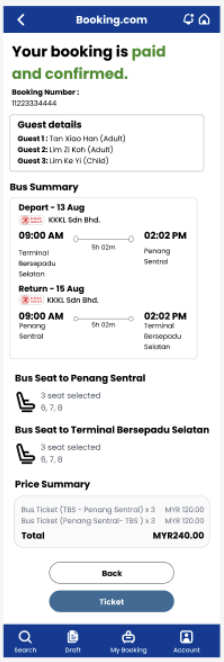
Detailed Bookin...



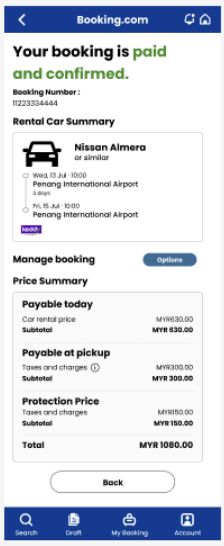
Detailed Bookin...



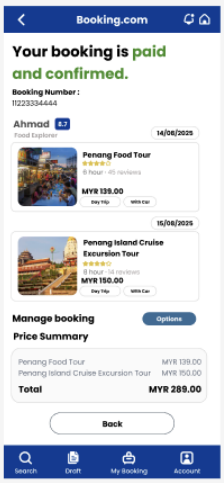
Detailed Bookin...



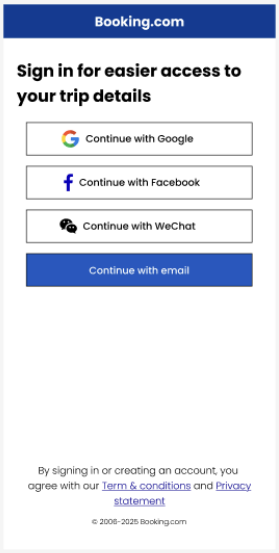
Detailed Bookin...



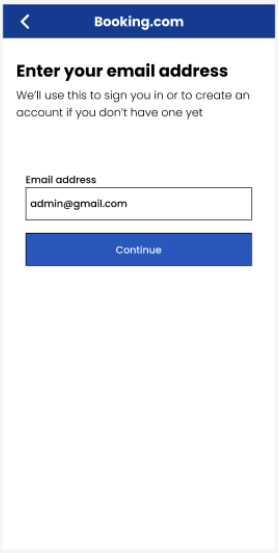
Detailed Bookin...



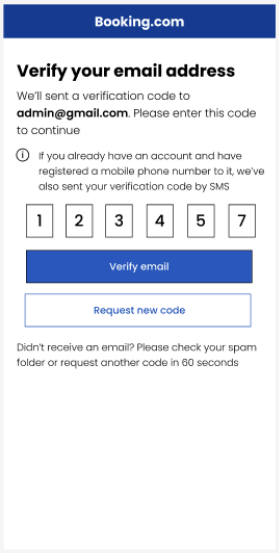
(A1) Login Page



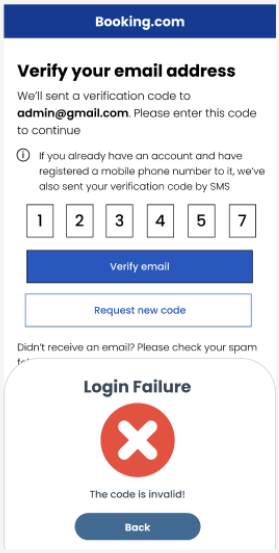
(A1) Email page



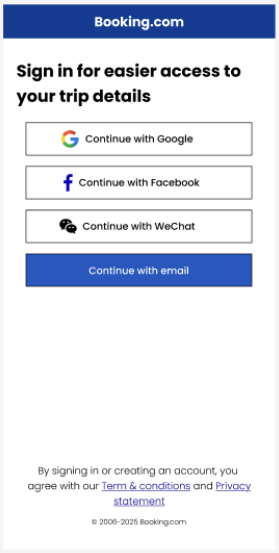
(A1) Verification page 1



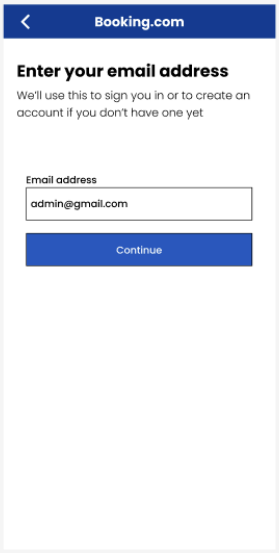
(A1) Verification page 2



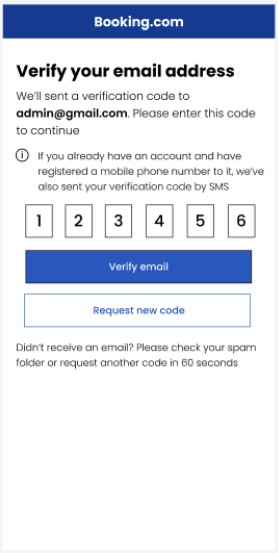
Login Page



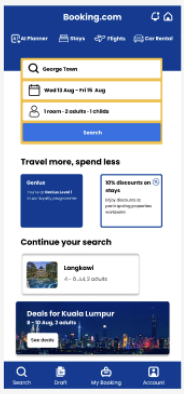
Email page



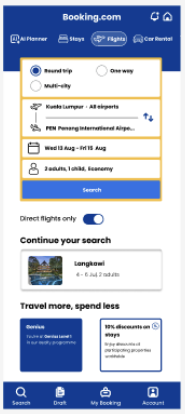
Verification page



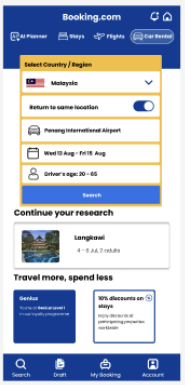
Home Page 1



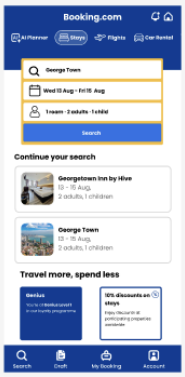
Flight HP



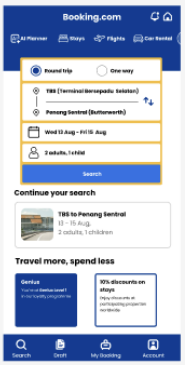
Car Rental HP



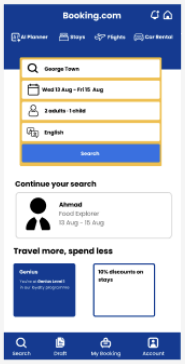
Manual Hote...



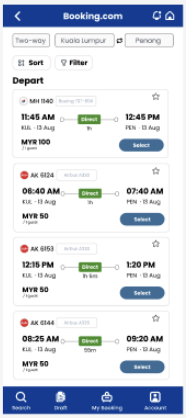
Bus Ticket HP



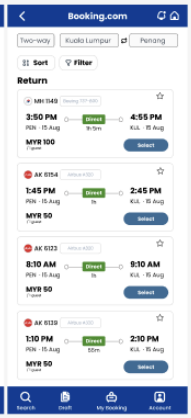
Tour Guide HP



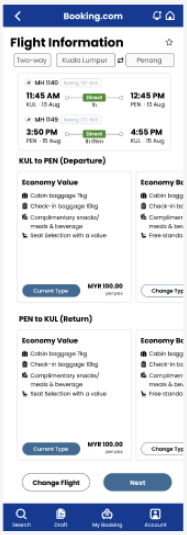
Flight Listing



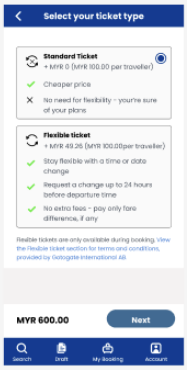
Flight Listing 2



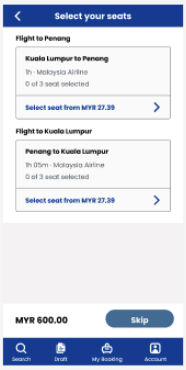
Flight Listing 3



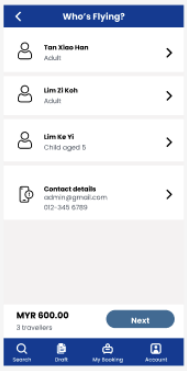
Flight Listing 4



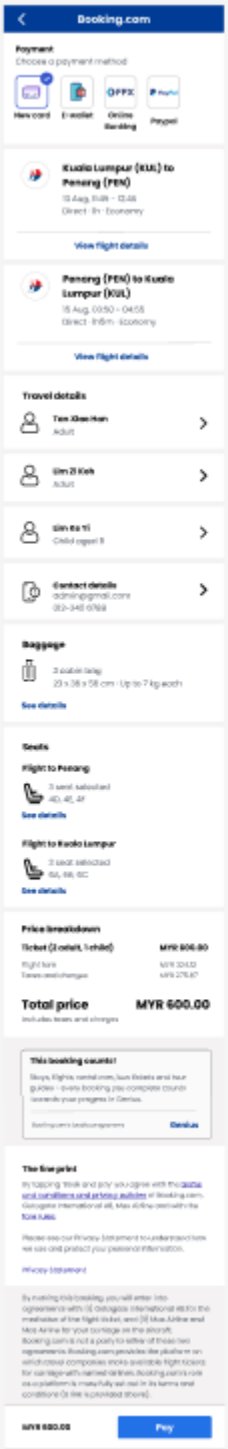
Flight Listing 5



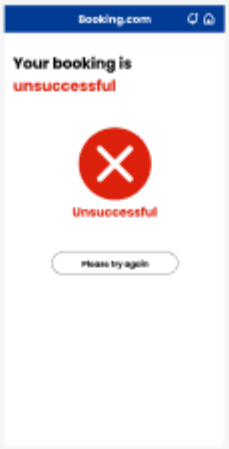
Flight Listing 6



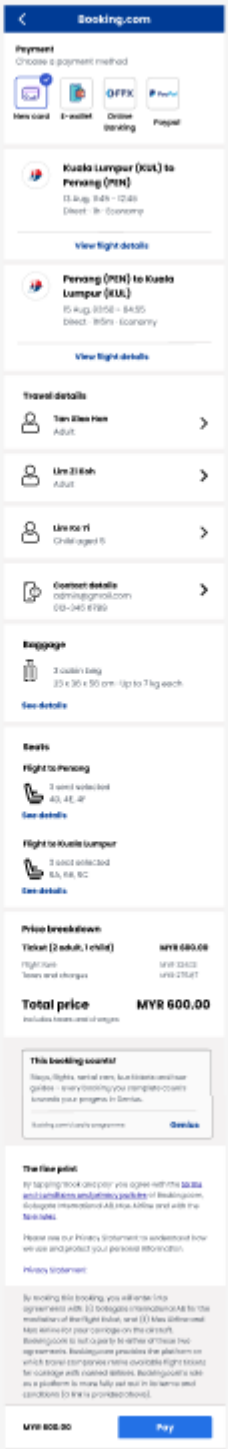
Flight Listing 7



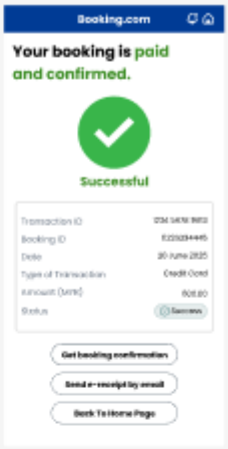
Flight Listing 8



Flight Listing 9



Flight Listing 10



Car Rental Lis...



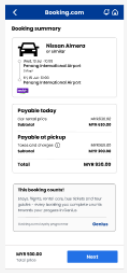
Car Rental Lis...



Car Rental Lis...



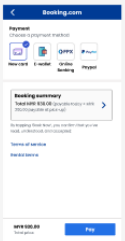
Car Rental Lis...



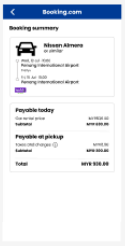
Car Rental Lis...



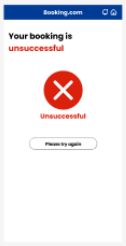
Car rental 6



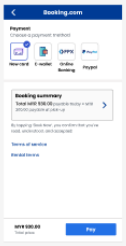
Car Rental Lis...



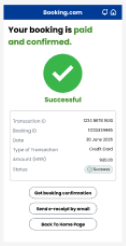
Car rental 7



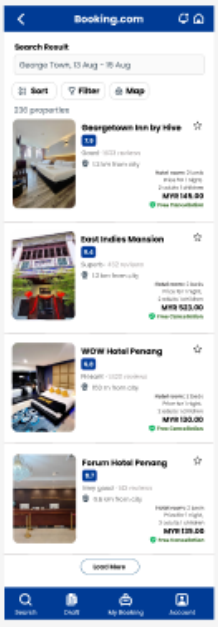
Car rental 8



Car rental 9



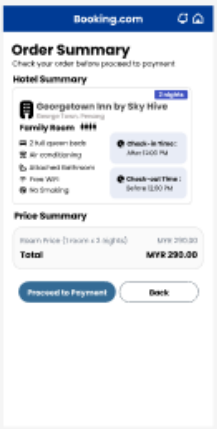
Hotel Listing



Property Inform...



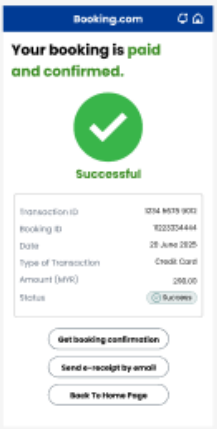
(M) Hotel Order ...



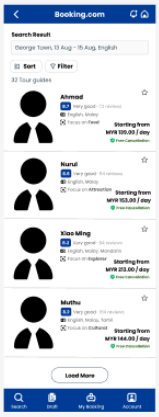
(M) Hotel Payme...



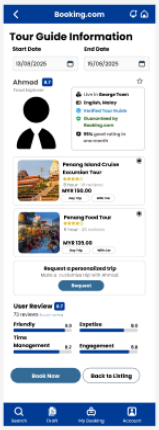
(M) Hotel Purch...



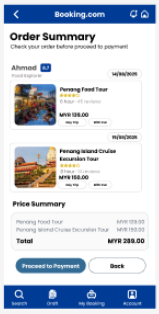
Tour Guide Listing



Tour Guide Inform...



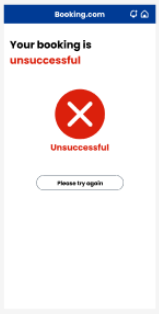
Tour Guide Order ...



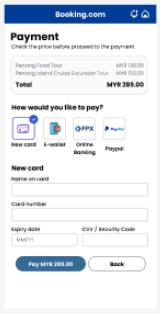
Tour Guide Payme...



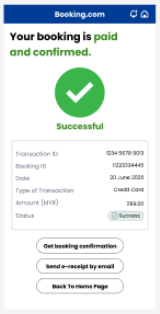
Tour Guide Purch...



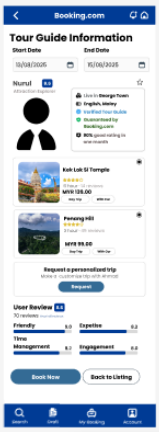
Tour Guide Payme...



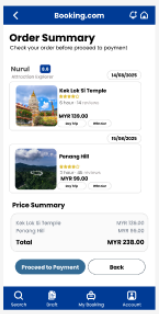
Tour Guide Purch...



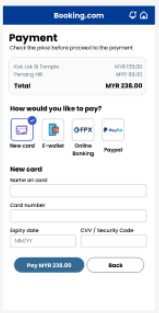
Tour Guide Inform...



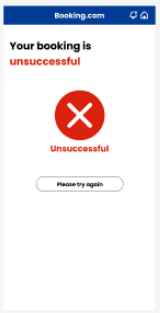
Tour Guide Order ...



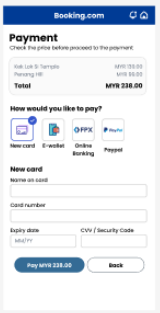
Tour Guide Payme...



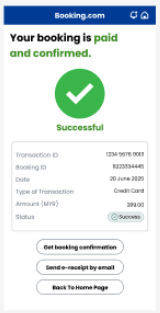
Tour Guide Purch...



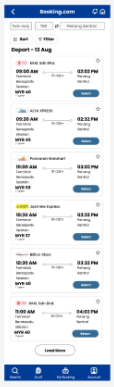
Tour Guide Payme...



Tour Guide Purch...



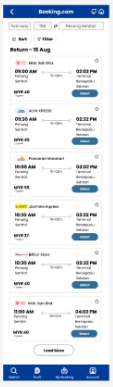
Bus Listing 1



Bus Listing 2



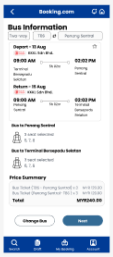
Bus Listing 3



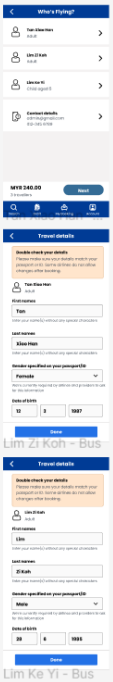
Bus Listing 4



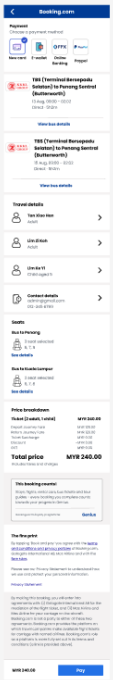
Bus Listing 5



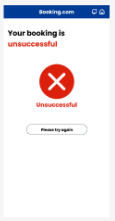
Bus Listing 6



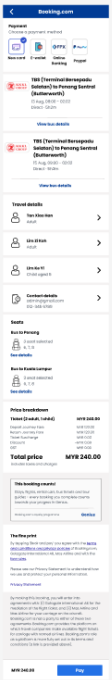
Bus Listing 7



Bus Listing 8



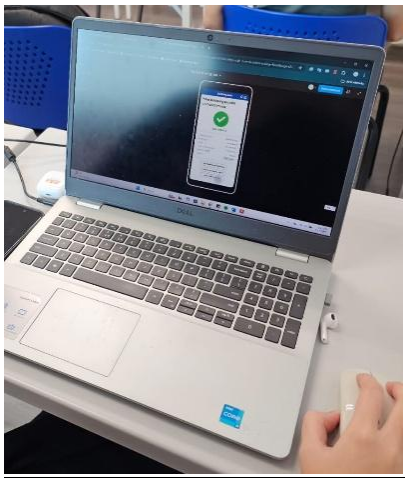
Bus Listing 9

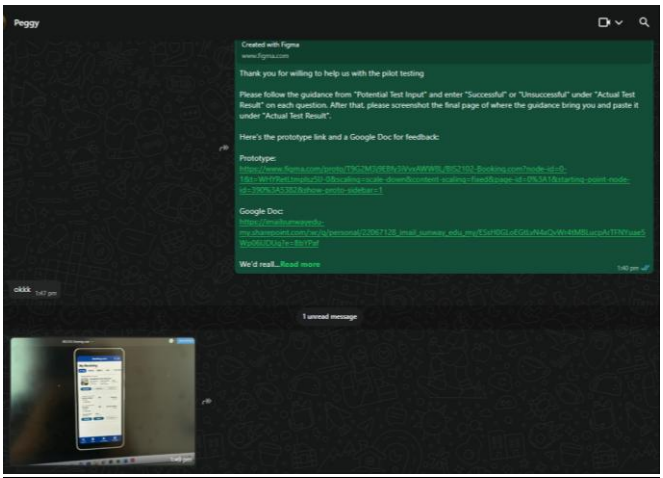


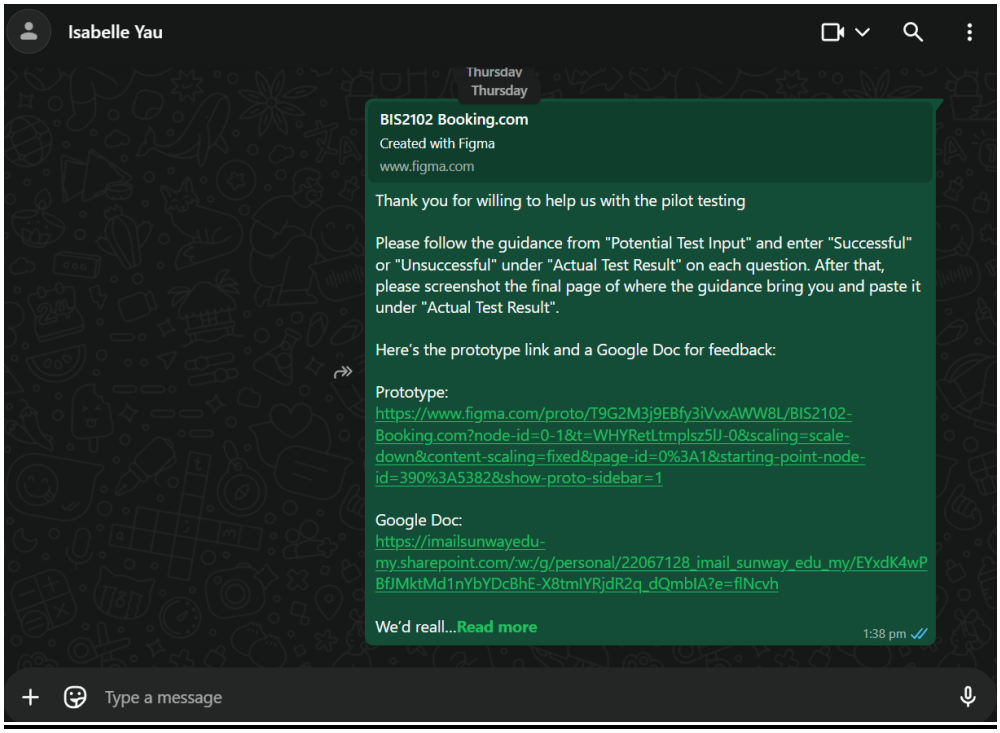
Bus Listing 10

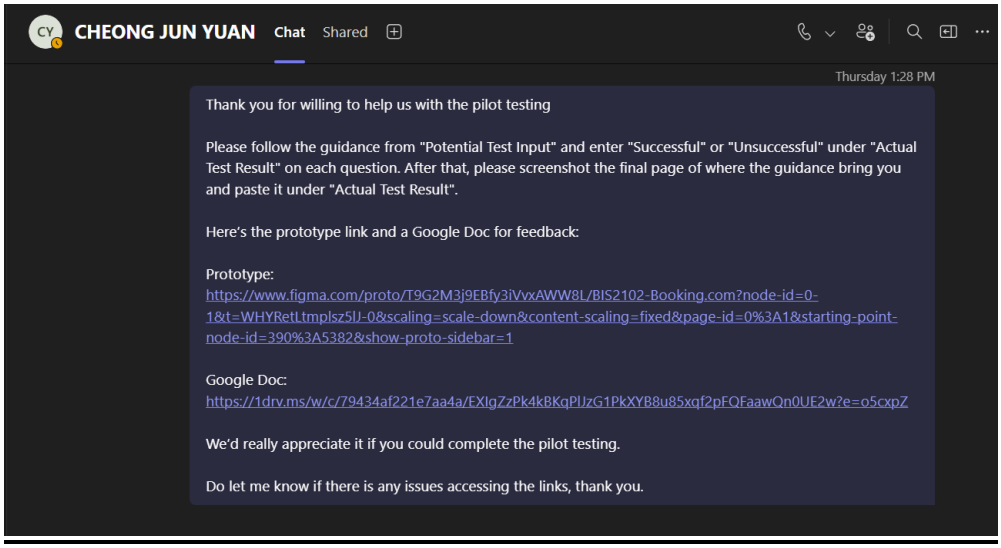


6.0 Appendix




Peggy
Thank you for willing to help us with the pilot testing
Please follow the guidance from "Potential Test Input" and enter "Successful" or "Unsuccessful" under "Actual Test Result" on each question. After that, please screenshot the final page of where the guidance bring you and paste it under "Actual Test Result".
Here's the prototype link and a Google Doc for feedback:
Prototype:
Google Doc:
We'd reall...Read more
okkk
1 unread message


Isabelle Yau
Thursday
Thursday
BIS2102 Booking.com
Created with Figma
www.figma.com
Thank you for willing to help us with the pilot testing
Please follow the guidance from "Potential Test Input" and enter "Successful" or "Unsuccessful" under "Actual Test Result" on each question. After that, please screenshot the final page of where the guidance bring you and paste it under "Actual Test Result".
Here's the prototype link and a Google Doc for feedback:
Prototype:
https://www.figma.com/proto/T9G2M3j9EBfy3iVvxAWW8L/BIS2102-Booking.com?node-id=0-1&t=WHYRetLtmplsz5lJ-0&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=390%3A5382&show-proto-sidebar=1
Google Doc:
https://imailsunwayedu-my.sharepoint.com/:w:/g/personal/22067128_imail_sunway_edu_my/EYxdK4wPBfJMktMd1nYbYDcBhE-X8tmlYRjdR2q_dQmbIA?e=flNcvh
We'd reall...Read more
1:38 pm
Type a message


CHEONG JUN YUAN
Chat
Shared
Thursday 1:28 PM
Thank you for willing to help us with the pilot testing
Please follow the guidance from "Potential Test Input" and enter "Successful" or "Unsuccessful" under "Actual Test Result" on each question. After that, please screenshot the final page of where the guidance bring you and paste it under "Actual Test Result".
Here's the prototype link and a Google Doc for feedback:
Prototype:
https://www.figma.com/proto/T9G2M3j9EBfy3iVvxAWW8L/BIS2102-Booking.com?node-id=0-1&t=WHYRetLtmplsz5lJ-0&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=390%3A5382&show-proto-sidebar=1
Google Doc:
https://1drv.ms/w/c/79434af221e7aa4a/EXlgZzPk4kBKqPIJzG1PkXYB8u85xqf2pFQFaawQn0UE2w?e=o5cxpZ
We'd really appreciate it if you could complete the pilot testing.
Do let me know if there is any issues accessing the links, thank you.